Screening for Coordinated Manipulation and Evaluating Exchange Rules in a Calibrated Market Simulator

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Abstract

Prosecuted stock-market manipulation is often carried out by groups of accounts trading in step, yet regulators have few labelled cases to train a detector and the order-level data to test one are rarely public. We ask which rules and how much inspection make such a ring give up. We state a label-free coincidence test with an exact circular-shift null per account, derive a power law in which the shift of the statistic falls like the inverse square root of the number of accounts; the law fits an agent-based market and yields a deterrence condition linking inspection capacity to the fine. A ring that picks its best strategy against each rule, in simulated markets matching a stock on the Ho Chi Minh exchange, is unaffected by the daily price band, loses most of the block value when off-book blocks are priced at a trailing mean, and is deterred by inspection only with enough capacity; in one market, minimum resting times and cancel fees do not reduce the gains of fixed manipulators and raise volatility. On real order flow from a public derivatives venue the exchangeability condition fails for more than a quarter of active accounts, and a planted ring of 12 quiet accounts is found only at about a fifth of the flow. On that exchange, the 2016 tick reform lowered the share of zero-return days by seven points, and a price-based rank puts 8 of 10 documented cases in the daily top ten on at least one day.

Keywords: market manipulation, market surveillance, agent-based model, limit order book, exchange rules, regulation

JEL Classification: G18 , G14 , C63 , D47 , K42

1 Introduction

A securities regulator that wants to find manipulation faces three problems at once. It rarely has labelled cases, because enforcement is slow and selective. It cannot easily evaluate a rule change before introducing it, because the behaviour of manipulators changes with the rule. And the data that would let an outside researcher test a method, orders tagged with an account, is held only by the exchange and the regulator. These problems are acute in emerging markets. In ten recent enforcement decisions of one regulator, the State Securities Commission of Vietnam, with exact dates (Appendix A) the manipulation was carried out with between 5 and 164 accounts (median 24.5), by trading continuously, or crossing orders among accounts, to create the appearance of supply and demand. Spoofing and layering, a focus of the detection literature on order-book data (Tuccella et al. 2021; Fabre and Challet 2025), are not what the decisions describe.

This paper asks two questions of a regulator in that position and answers them with a combination of theory, simulation and public data. *Can a ring of accounts be found without labelled cases?* And *which rules and how much inspection capacity make a ring that adapts to them give up?* For the first we study a label-free coincidence test: each account's submissions are compared with the aggregate of all other accounts, and the comparison is calibrated by circularly shifting the account's own series, which gives an exact p-value under a stated exchangeability condition. For the second we let a ring choose its best strategy against each rule, from a fixed set, in families of simulated markets that match published and measured moments of a mid-cap stock on the Ho Chi Minh Stock Exchange (HOSE), and we relate the answer to closed-form conditions.

We make six contributions. (i) We state the test, prove that it controls false alarms exactly per account (Proposition 1), and show that ranking many accounts at once cannot inherit that guarantee because of the resolution of the shift distribution (Proposition 2). (ii) We derive a law for its power, in which the shift of the statistic falls like $N^{-1/2}$ in the number of accounts and rises with the number of co-active members (Proposition 3), verify it by Monte Carlo and show that it reproduces the dependence on the number of accounts in the agent-based market, with a fitted exponent of 0.32 against 0.5 in the theory; we show why netting buys against sells protects against honest market-making desks (Proposition 4). (iii) We turn the law into a deterrence condition for a regulator with a fixed inspection capacity and a fine proportional to the gain (Proposition 5), and derive and test a law for a rule that prices off-book blocks at a trailing mean (Proposition 6). (iv) We evaluate rules by the best response of a ring over two families of six calibrated markets, with the effect of rules on honest participants (Section 6). (v) We stress-test the statistic on real account-level order flow from a public venue, where it fails the exchangeability condition for a large share of accounts and detects a planted ring less often than in the simulator (Section 7.1), and we check what public data from that exchange can say: documented enforcement cases (Section 7) and a natural experiment on the tick size (Section 6.8), and we use public cryptocurrency pump events, which are not from that exchange, to see how weak the preparation phase of a pump is in tick data (Section 7.3). (vi) We release the code and all result tables.

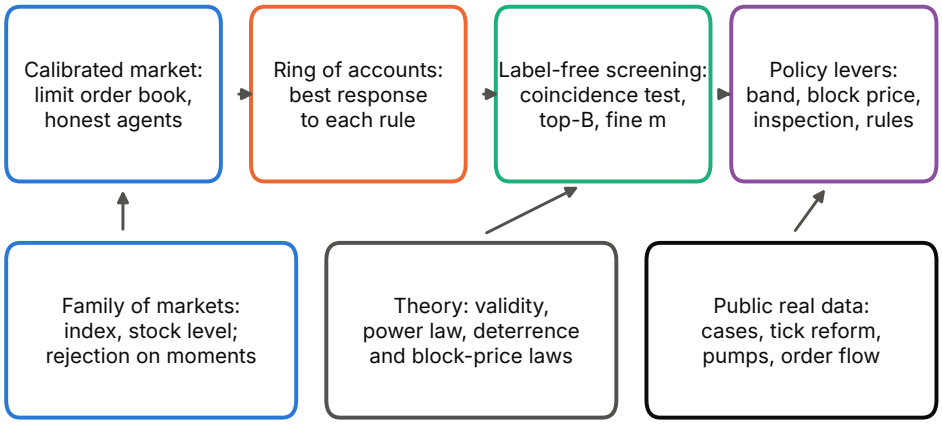


Fig. 1 Overview of the study. A calibrated market with honest agents and groups, a ring that chooses its best strategy against each rule, a label-free screening step with a fixed inspection capacity and a fine, and the policy levers that are compared; the evidence comes from families of calibrated markets, from theory checked by simulation, and from public real data

The paper does not claim that the detector works on real order data from a stock exchange. Its manipulators and features are written by us, a supervised model trained in the simulator does better than the label-free test in every simulated scenario, the real account-level data come from a different kind of market, and the checks on public exchange data are at the level of the stock, not of the account. The contribution is a calibrated screening tool with a measured failure profile, a theory that links its power to deterrence capacity, and an evaluation method for rules that does not assume manipulators stand still. Section 2 places the paper among the related work, Section 3 sets up the market and the regulator's problem, Section 4 gives the method and its theory, Sections 5–7 report the experiments, and Section 8 discusses the impact for policy and the limits. Figure 1 gives an overview.

2 Related work and position of this paper

Four literatures meet in the problem we study: detecting manipulation, finding coordination among traders, testing coincidence between event series, and using simulated markets and enforcement theory to choose rules. For each we say what it provides, what it assumes, and what is left for this paper; Table 1 summarises.

Detection with labelled events or order-book data.

Most detection work targets a manipulation that leaves a public trace or a label. For pump-and-dump schemes organised on messaging channels the announcement itself labels the event: Xu and Livshits (2019) study 412 pumps announced on Telegram (June 2018 to February 2019) and predict which coin will be pumped; Hu et al. (2023)

model 709 events from 2019 to 2022 with a sequence network over channel histories; La Morgia et al. (2020) propose a real-time detector and La Morgia et al. (2023) reach an F1 score of 94.5% with detection 25 seconds after a pump starts, on a data set of about 900 events; Dhawan and Putniņš (2023) measure average price distortions of 65% on 355 cases; Kamps and Kleinberg (2018) define criteria and apply anomaly detection to trade data. For spoofing, Tuccella et al. (2021) train a recurrent network on granular order-book data for early detection, and Fabre and Challet (2025) predict the distribution of mid-price moves from Hawkes-process order-flow variables and find that 31% of the large orders in a four-day sample (4 to 7 December 2024) of Level-3 data could spoof. Wash trading is detected from structure: suspiciously matched orders and trader collusions in directed trader graphs on stock data (Cao et al. 2016), accounts and trading structures that meet legal definitions on two blockchain exchanges (Victor and Weintraud 2021), and statistical regularities of reported volume, which put wash trading above 70% of reported volume on unregulated exchanges (Cong et al. 2023). Most of these methods need labels (announcements, cases) or counterparty-level records, the exceptions being statistical regularities of reported volume (Cong et al. 2023) and anomaly detection on trade data (Kamps and Kleinberg 2018), and the spoofing and pump settings are not those of the Vietnamese cases, in which groups of accounts trade among themselves or continuously for months (Appendix A). We use a pump data set only to measure how weak the preparation phase of a pump is in public tick data (Section 7.3); our detector uses neither labels nor counterparties, only the timing and side of submissions.

Coordination among traders.

Closer to our setting, collusion has been sought in trade networks. Palshikar and Apte (2008) cluster the trade graph and combine evidence from several algorithms; Sun et al. (2012) test the correlation between the degree and strength of traders in networks of eight manipulated and 44 non-manipulated stocks against randomised networks; Jaeger et al. (2025) link insiders by the temporal similarity of their reported trades in 2.9 million trades reported to the SEC, motivated precisely by the lack of labels. Saavedra et al. (2011) find that day traders who trade more in step with others are less likely to lose money, which suggests that synchrony is not in itself a sign of misconduct. That finding is the reason for our emphasis on honest groups (market-making desks, order-splitting funds, bot fleets) and for the netting of buys against sells (Proposition 4), and the real order flow of Section 7.1 is consistent with it in that a large share of active accounts coincide with the others beyond chance; we do not test whether synchronised accounts are more profitable. Relative to this strand we compare one account with the *aggregate* of all others rather than pairs, which pools evidence when a ring fragments its trading over many accounts, and we attach an exact null to the comparison.

Coincidence tests with surrogates.

Testing whether events in two series coincide more than chance without a model for either series is a long-standing problem in neuroscience. Grün (2009) show that nonstationary rates and non-Poisson structure generate false positives unless they are

part of the null; Louis et al. (2010) build surrogates that keep rates and inter-event intervals; Amarasingham et al. (2012) review jitter resampling and exact tests for repeated fine-scale patterns; Platkiewicz et al. (2017) show that, for processes without temporal structure, interval jitter gives an exact test while spike-centred jitter can inflate false positives; Liang et al. (2025) give finite-sample false-positive guarantees for time-shift tests that allow dependence within the series. We transfer the circular-shift version to accounts, add netting, take the maximum over tolerances inside one shift distribution, and prove two properties that matter for a regulator: the resolution limit of the discrete p-value across many accounts (Proposition 2) and a power law in the number of accounts (Proposition 3). The warning of Grün (2009) is the main lesson of our real-data test: common drivers make the null false for a large share of the active accounts.

Simulated markets for rules and for surveillance.

Open simulators exist (Byrd et al. 2020), and their realism is a recognised issue: Vyetrenko et al. (2020) compare simulated and real limit order books on stylised facts and find more realistic behaviour when the fundamental comes from historical data, and Platt and Gebbie (2018) find that microstructure parameters of agent-based models can be calibrated but behavioural parameters remain problematic. Both point to the design we adopt, a family of markets matching the observable moments instead of a single tuned market (Section 4.7). Agent-based models have been used to study rules: tick size, where quality weakens as the tick grows and is lowest for small caps (Zhao et al. 2020); price limits, which cause volatility spillover and, when upper limits are imposed, delay price discovery (Zhang et al. 2016); and minimum resting times, circuit breakers, cancellation fees and taxes, which trade stability against resilience in a market with high-frequency traders (Leal and Napoletano 2019). For manipulation specifically, Wang and Wellman (2020) generate order streams in which a manipulator mimics a market maker, and Wang et al. (2018) analyse a cloaking rule against a spoofing exploiter by empirical game-theoretic analysis; Cartea et al. (2020) show in a control model that a larger fine makes spoofing less attractive until it disappears. These studies concern single manipulators and generic markets. We study a ring, a specific venue, and the best response of the manipulator to each rule in a family of calibrated markets, add closed-form conditions (deterrence capacity, block-price law), and verify the mechanism ex post.

Enforcement economics and Vietnamese evidence.

Deterrence depends on the probability of detection and the size of the sanction (Becker 1968; Polinsky and Shavell 2000); inspection games model inspection against an adversary (Avenhaus et al. 2002), and double-oracle methods (McMahan et al. 2003) and their multiagent generalisation (Lanctot et al. 2017), which iterate best responses to mixtures of opponent policies, are a natural next step for the interaction between an adaptive ring and an adaptive regulator. Aggarwal and Wu (2006) use SEC cases to show that manipulation raises volatility, liquidity and returns during the episode, with prices falling afterwards; Comerton-Forde and Putniņš (2014) estimate that about 1% of closing prices are manipulated, that only a small fraction is detected and prosecuted,

strand	typical input	labels needed	account-level data	error control	policy evaluation
pump-and-dump detection	public tick data and announcements	yes	no	empirical	no
spoofing detection	order book, orders	yes	no (one trader)	empirical	little
wash-trading detection	counterparty-level trades or reported volume	rules or none	yes	rules, empirical	no
trader-network collusion	trade graphs, filings	some	yes	randomised networks	no
time-shift and jitter tests	event series (neuroscience)	no	not applicable	exact or finite-sample	no
agent-based rule studies	simulated markets	no	simulated	not applicable	yes, fixed agents
this paper	order submissions (time, side)	no	yes (not public here)	exact per account; ranking in practice	yes, best response over calibrated markets

Table 1 Positioning of the paper among the strands of related work.

and that the regulator's budget has a strong effect on both manipulation and detection, which is the empirical counterpart of our capacity result (Proposition 5); Khwaja and Mian (2005) study price manipulation by intermediaries in an emerging stock market, and Nguyen et al. (2017) insider trading around stock splits in an emerging market. For Vietnam, Nguyen and Vo (2024) study herding across COVID-19 periods, and Vo and Doan (2023) find that the 2016 tick reduction lowered trading costs on the Ho Chi Minh Stock Exchange, a result we use as an external check (Section 6.8). None of these studies has account-level manipulation data for Vietnam, and neither do we.

Position.

The gap this paper addresses is the combination of the strands in the last row: a screening statistic that needs neither labels nor counterparties, whose error properties and power are stated and checked, together with a policy evaluation in which the manipulator adapts and the market is not a single draw. Our claims are limited accordingly. Components are not new (shift tests, adversarial simulation, enforcement economics); the contribution is their calibrated combination, the theory that connects detection power to deterrence capacity and to the effect of a block-price rule, and the measurement of where the combination fails, on simulated honest groups and on real account-level order flow.

3 Setting: a calibrated market, a ring and a regulator

3.1 The market and the manipulators

Market.

A limit order book with price-time priority and self-trade prevention; one step is about one trading minute and a day has 240 steps. Honest agents are noise traders, market makers that re-quote two levels per side and shift with displayed depth imbalance, fundamental traders that follow a noisy latent value, capped momentum traders, depth-imbalance traders, and institutions that slice a parent order into market orders.

Manipulators.

A *ring* of K accounts (3 to 40) runs cycles of accumulation by passive buys, a push by near-simultaneous aggressive buys from several members with some cross trades, and distribution by passive sells of the stock it holds; no member sells short. Orders are spread over members; a jitter parameter spreads the members' actions in time. Outside its campaign each account trades as a noise trader. The ring's benefit is a block held beforehand (3000 lots, an assumption) and sold off-book at the end of each push, valued at the achieved price displacement, plus trading profit. Single-account spoofers, pump-and-dump agents and wash pairs are also implemented.

Honest groups that act in step.

A market-making firm with three to five desks that re-quote together; a fund that splits parent orders across 6 to 12 sub-accounts in the same step; and a fleet of bots that react to one public price signal with small latency differences.

Rules.

The exchange can apply a tick size, minimum resting time, cancel fee, circuit breaker, daily price band, and a reference-price rule for off-book blocks (the last price or a trailing mean).

3.2 The regulator's problem

Let a policy $\pi = (\text{band}, L, B, \text{order-level rules})$ be a choice of the daily price band, the block reference window L (Section 4.6), the inspection capacity B per window and the order-level rules. A market φ is a draw of the honest participants (numbers and activity of noise, fundamental and momentum traders and market makers). The ring chooses a strategy θ from the set Θ of Table 2 (size, orders per push, members per push, push and accumulation probabilities, jitter, cross-trade share, phase lengths) to maximise its net benefit $U(\theta; \pi, \varphi)$ of (5); the regulator observes the ring's value $V(\pi; \varphi) = \sup_\theta U$ and the cost $C(\pi; \varphi)$ to honest participants, measured by relative changes of the spread, depth and volatility of the honest-only market. The regulator is the leader and the ring the follower of a Stackelberg game; we do not claim an equilibrium for repeated play, in which the regulator would also adapt (Section 8). Inspection of the B top-ranked accounts uses the signed test of Section 4.1 on windows of 300 steps. A ring is *caught* if one member is inspected in some window of the episode, on the assumption that finding one member exposes the others, as account linking does in practice, and a caught ring pays m times its positive gain, with $m = 5$ for individuals and $m = 10$ for organisations (the caps of Vietnamese administrative fines, Appendix B).

3.3 Calibration to a HOSE stock

Moments.

Prices are in ticks of the Ho Chi Minh Stock Exchange (HOSE): a stock at 25,000 Vietnamese dong (VND) with a 50 VND tick has a relative tick of 20 bp, and the daily

parameter	range	meaning
K	3–40	number of ring accounts
q	1–6	order size per push order (lots)
m_{push}	1–4	members that push together
p_{push}, p_{acc}	0.2–1.0, 0.1–0.8	probability that a member acts in a push step, in an accumulation step
jitter	0–40 steps	spread of members' actions in time
cross share	0–0.5	share of cross trades among members
$\ell_{\text{acc}}, \ell_{\text{push}}, \ell_{\text{dist}}$	20–100, 10–50, 30–80 steps	phase lengths
cool-down	30–120 steps	pause between cycles

Table 2 Ring strategy space Θ (11 parameters).

band is ±7% of the previous close. Rules and statistics come from broker summaries and press (Appendix B); the exchange rulebooks could not be fetched. We add measurements from public datasets (Table 3). Table 4 compares the first preset with the real moments. Cancel share, relative tick and daily volatility are matched; the spread (3 ticks against one tick in 73% of futures quotes), the tails (excess kurtosis 4–5 daily and 40 at one minute) and the serial correlation of daily returns are not: simulated daily returns trend, with a lag-1 autocorrelation of 0.49 at index level and 0.60 at stock level against −0.03 for VN30 and 0.03 on average for HOSE stocks (Figure 2c). The trending comes from a slow adjustment of the price to the latent value; a family of markets in which it is calibrated away is analysed in Section 6.3.

Index level and stock level.

The 1.16% target is the volatility of the VN30 *index* (the 30 largest HOSE stocks). A single stock is more volatile: the median HOSE stock has a daily volatility of 2.46% since 2022, and the stocks in the enforcement cases had a median of 1.75% in the year before the manipulation (nine of the ten, those with earlier price history), with a share of zero-return days of 0.13 against 0.12 for HOSE stocks (Section 7.2). The case stocks are therefore ordinary HOSE stocks, not unusually illiquid ones, and the daily band was reached in their manipulation periods (median five days within 0.5 points of the ceiling; one stock, with an assumed 7% band, for 12 days in a row). We therefore build two families of calibrated markets: index level (daily volatility 0.9–1.4%) and stock level (1.8–2.8%).

4 A label-free screening statistic and its theory

This section states the test, proves when it controls false alarms, gives a law for how its power depends on the number of accounts and on the size of a ring, shows why netting buys against sells protects against honest groups, and turns the power law into a deterrence condition for a regulator with a fixed inspection capacity. Section 5 checks every statement by simulation with the real implementation, and compares the power law with the agent-based market.

4.1 The coincidence statistic

Take a window of T steps and N accounts. Let $x_i \in \mathbb{R}^T$ be account i's signed submissions, $x_i[t]$ = buy orders minus sell orders submitted at step t, and let

data	content	used for
Daily prices of 1,551 Vietnamese stocks, 2015–2025	open, high, low, close, volume	stock-level detection checks (Section 7), tick-reform experiment (Section 6.8), case statistics
Exchange metadata	HOSE, Hanoi Stock Exchange (HNX) or Unlisted Public Company Market (UPCoM) for each ticker	control group for the tick reform; band of each case stock
VN-Index (from 2000), VN30 (2012) and VN100 (2014) bars	daily and 30-minute bars	index-level volatility target
1.56 million VN30 futures ticks, 2024–25	price, best bid and ask	spread, tick activity (not matched)
10 documented enforcement cases	stock, exact period, accounts, decision	weak labels for detection checks
701 Telegram-organised pumps	tick trades around the pump, no accounts	accumulation before pumps, early-warning limits
Published rules and statistics	tick, band, lot, cancel share, retail share, fine cap	simulator rules and moments
Order submissions of 1,909 accounts, Hyperliquid SOL perpetual, 23.6 minutes	order, account, side, time	calibration and planted ring on real accounts (Section 7.1)
Fills of three Hyperliquid perpetuals, 91 minutes	account address, side, millisecond time	check of the mirror effect of executed trades

Table 3 Data used. Only the last two rows contain account identifiers (pseudonymous addresses on a crypto derivatives venue); no Vietnamese account-level data are used.

moment	default preset	first accepted index-level market	real	source
cancel share of orders	0.321	0.318	0.318	HOSE, December 2020 (press)
relative tick (bp)	21	19.8	20	derived from the tick table
daily volatility, index level (%)	1.14	1.05	1.16	VN30, 2023 onward
daily volatility, typical stock (%)	–	–	2.46	HOSE median stock, 2022 onward
retail share of value	0.72	0.717	0.75–0.82	press estimates, 2024–25
spread: share of quotes at one tick	0	0	0.73	VN30F1M, 2024–25
lag-1 autocorrelation of daily returns	–	0.49 (stock level: 0.60)	−0.03 (VN30), 0.03 (HOSE stocks)	2022–25

Table 4 Calibration of the simulator. The default preset is the single market used first (values from its calibration); the second column is re-measured over 16 honest episodes of the first accepted market of the index-level family. The last two rows are not matched; the stock-level family targets the volatility of a typical stock

$y_{-i} = \sum_{j \neq i} x_j$ be the flow of all other accounts. For a tolerance $\tau \in \mathcal{T}$ let $\tilde{y}^{\tau}_{-i}[t] = \sum_{|d| \leq \tau} y_{-i}[(t+d) \bmod T]$ (a circular box filter). For a shift $k \in \{0, \dots, T-1\}$ define the circular cross-correlation and the standardised statistic

$$C_i^{\tau}(k) = \sum_{t=0}^{T-1} x_i[(t-k) \bmod T]\, \tilde{y}^{\tau}_{-i}[t], \qquad Z_i(k) = \max_{\tau \in \mathcal{T}} \frac{C_i^{\tau}(k) - \bar{C}_i^{\tau}}{s_i^{\tau}}, \tag{1}$$

where $\bar{C}_i^{\tau}$ and s_i^{τ} are the mean and standard deviation of $k \mapsto C_i^{\tau}(k)$ over the T shifts. The statistic is large when account i submits on the same side as the others at the same moments. The observed value is $Z_i(0)$; the circular-shift p-value is

$$p_i = \frac{1}{T}\, \#\{k \in \{0, \dots, T-1\} : \, Z_i(k) \geq Z_i(0)\} \in [1/T, 1]. \tag{2}$$

All T shifts are computed by the fast Fourier transform, so one window costs $O(NT \log T)$ per tolerance; we use $T = 300$ and $\mathcal{T} = \{1, 5, 15\}$. The *unsigned* variant keeps buys and sells as two channels $x_i^{\mathrm{b}}, x_i^{\mathrm{s}} \geq 0$ and adds their correlations before

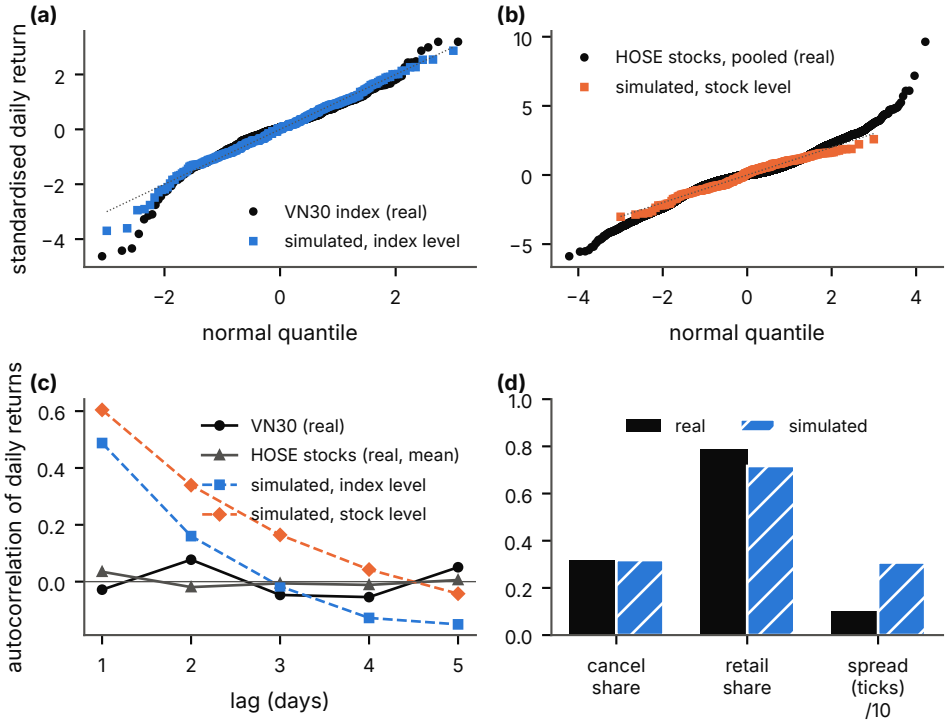


Fig. 2 Calibration of the simulated markets against real data. (a) Standardised daily returns of the VN30 index since 2023 and of the first accepted index-level market, against normal quantiles. (b) Pooled standardised daily returns of 378 HOSE stocks since 2022 and of the first accepted stock-level market. (c) Autocorrelation of daily returns at lags 1 to 5: VN30 index, mean over HOSE stocks, and the first accepted market of each simulated family (pooled over episodes). (d) Cancel share, retail share (midpoint of the published range 0.75–0.82) and spread in ticks divided by 10 (VN30 futures: one tick for 73% of quotes); the simulated spread is not matched

standardising. In use, the B accounts with the largest $Z_i(0)$ in a window are inspected ("top-B"), or accounts with $p_i \le \alpha$ are flagged.

4.2 Exact false-alarm control and its limits

Proposition 1 (validity) *Suppose that under the null hypothesis for account i the pair (x_i, y_{-i}) is invariant in law to a cyclic shift of x_i alone: $(g\,x_i,\, y_{-i}) \overset{d}{=} (x_i,\, y_{-i})$ for every cyclic shift g. Then $\Pr(p_i \le \alpha) \le \alpha$ for every $\alpha \in [0,1]$.*

Proof Write $Z_k = Z_i(k)$ and $R_m = \#\{k : Z_k \ge Z_m\}$. Shifting x_i by g^j shifts the sequence $k \mapsto C_i^\tau(k)$ cyclically, so its mean and standard deviation over k do not change, and $Z_k(x, y) = Z_0(g^k x, y)$. Hence $R_m(x, y) = R_0(g^m x, y)$ and by invariance $R_0 \overset{d}{=} R_m$ for every m. For any real vector (Z_k) and any $r > 0$, at most r indices m satisfy $R_m \le r$: if S is the set of such

indices and $m^* \in S$ has the smallest Z_{m^*}, then every $m \in S$ has $Z_m \geq Z_{m^*}$, so $|S| \leq R_{m^*} \leq r$. Therefore $\Pr(R_0 \leq \alpha T) = \frac{1}{T}\sum_m \Pr(R_m \leq \alpha T) = \frac{1}{T}\mathbb{E}\,\#\{m : R_m \leq \alpha T\} \leq \alpha$. □

The maximum over tolerances is taken inside $Z_i(k)$ for the same k, so it does not break the argument. This is the standard randomisation-test argument (see Liang et al. 2025, for time-shift tests on dependent series), and its condition is strong: it fails when account i and the others share a trend or a burst inside the window, and when honest accounts react to the same event. Honest groups that act in step violate it; Section 5.3 shows how much and how netting removes most of it. Two consequences of the discreteness of p_i matter for a regulator.

Proposition 2 (inspection budget and resolution) *Let H be a set of accounts for which the condition of Proposition 1 holds, with arbitrary dependence between them. (a) For any α, $\mathbb{E}\,\#\{i \in H : p_i \leq \alpha\} \leq |H|\alpha$; flagging at the threshold $\alpha = \lambda/N$ gives at most λ expected false flags per window, and at most λW over W windows. (b) Always $p_i \geq 1/T$. Hence flagging with a family-wise error rate α_0 by Bonferroni ($p_i \leq \alpha_0/N$) is possible only if $T \geq N/\alpha_0$, that is for at most $N_{\max} = \alpha_0 T$ accounts.*

Proof (a) is linearity of expectation applied to Proposition 1; no independence is needed. (b) p_i counts the shift $k = 0$ itself. □

With $T = 300$ shifts and $\alpha_0 = 0.05$ the Bonferroni rule can handle only 15 accounts, while a listed stock has hundreds of active accounts in a window of 300 minutes. Control with exact guarantees therefore needs either a finer resampling (more shifts than steps, as jitter surrogates provide) or ranking without a formal error bound. We use ranking (top-B), report the observed false-alarm rate, and treat the formal guarantee as per-account rather than per-family.

4.3 Power: a law in the number of accounts and the size of the ring

To see how detection depends on the market and on the ring, consider a Gaussian-flow model. Honest accounts have $x_i = e_i$ with $e_i[t]$ independent $\mathcal{N}(0, \sigma^2)$. A set of k *co-active* accounts (a ring, or the members that push together) each add the same signal, $x_i = e_i + b\,s$ for $i \leq k$, with $s[t] \in \{0, 1\}$ a campaign indicator and $\eta = b^2\mathrm{Var}(s)/\sigma^2$ the signal-to-noise ratio of one member.

Proposition 3 (signal index) *In this model the population correlation between a co-active member's flow and the flow of all other accounts is*

$$\rho_{N,k}(\eta) = \frac{(k-1)\,\eta}{\sqrt{(1+\eta)\,\big[(N-1) + (k-1)^2\eta\big]}}, \tag{3}$$

and an honest account has correlation 0. For a single tolerance τ and a window of T steps the statistic satisfies $Z_i(0) \approx \sqrt{T_{\mathrm{eff}}}\,\rho + \xi$ with ξ standard normal, where $T_{\mathrm{eff}} \approx T/(2\tau+1)$ for white flows. Consequently: (i) ρ increases in k and in η and saturates at $\sqrt{\eta/(1+\eta)}$ as $k \to \infty$; (ii) for $N \gg (k-1)^2\eta$, $\rho \approx (k-1)\eta/\sqrt{(1+\eta)N}$, so the shift of the statistic falls like

$N^{-1/2}$*; (iii) to keep a given power as* N *grows the number of co-active accounts must grow like* $\sqrt{N}$*: the detectable size is* $k-1 \gtrsim (t+z_\beta)\sqrt{(1+\eta)N}/(\eta\sqrt{T_{\text{eff}}})$*, where* t *is the critical value and* z_β *the normal quantile of the target power.*

Proof Let $v_s = \mathrm{Var}(s)$. For a member i and $y = y_{-i}$: $\mathrm{Cov}(x_i, y) = (k-1)b^2 v_s = (k-1)\eta\sigma^2$, $\mathrm{Var}(x_i) = \sigma^2(1+\eta)$, and $\mathrm{Var}(y) = (N-1)\sigma^2 + (k-1)^2 b^2 v_s = \sigma^2[(N-1) + (k-1)^2\eta]$, because the $N-1$ other noise terms are independent and the $k-1$ other members share the signal. Dividing gives (3). For honest i, e_i is independent of y. Standardising $C(k)$ across shifts converts $T\,\widehat{\mathrm{cov}}(x_i, \tilde{y})$ into $\sqrt{T}$ times the sample correlation when x and $\tilde{y}$ are white, and a box filter of width $2\tau+1$ divides the effective number of independent samples by $2\tau+1$; the sample correlation is asymptotically $\mathcal{N}(\rho, 1/T_{\text{eff}})$ for small ρ. (i)–(iii) follow from (3) by differentiation and by solving $\rho\sqrt{T_{\text{eff}}} = t + z_\beta$. □

Proposition 3 is an approximation, not an exact result (the sample-correlation law is asymptotic and the maximum over tolerances adds a positive bias to $Z_i(0)$ in both hypotheses); Section 5.2 checks the linear dependence on ρ and the $N^{-1/2}$ law in simulation. The predicted probability that a member is among the B top-ranked accounts of N follows from the null distribution F_0 of $Z_i(0)$: with $t_B = F_0^{-1}(1-B/N)$ and a mean shift $S = c\rho$,

$$q_B = 1 - F_0(t_B - S) \;\approx\; \Phi\big(S - \Phi^{-1}(1-B/N)\big), \tag{4}$$

and a ring of K accounts has a member among the top B with probability about $1-(1-q_B)^K$ if members are treated as independent (they are positively dependent, so the approximation overstates detection at small K; Section 5.2 quantifies this).

Heterogeneous accounts.

If honest accounts have different noise variances σ_j^2 and member i has σ_i^2, the same calculation gives ρ as in (3) with η replaced by $\eta_i = b^2\mathrm{Var}(s)/\sigma_i^2$ and with $N-1$ replaced by the *effective number of accounts* $N_{\text{eff},i} = \sum_{j\neq i}\sigma_j^2/\sigma_i^2$, because $\mathrm{Var}(y_{-i}) = \sum_{j\neq i}\sigma_j^2 + (k-1)^2 b^2 \mathrm{Var}(s)$ and $\mathrm{Var}(x_i) = \sigma_i^2(1+\eta_i)$. A quiet account in a market whose aggregate flow is dominated by a few loud accounts therefore sees an N_{eff} far above the number of accounts, and its ring is correspondingly hard to see; Section 7.1 measures $N_{\text{eff}} \approx 10^6$ against a median of 468 active accounts on a real order flow.

4.4 Honest groups and the netting of buys against sells

Market-making desks re-quote on both sides at the same moments. Suppose desk i submits $b_i[t] = u[t] + \beta_i[t]$ buy and $s_i[t] = u[t] + \sigma_i[t]$ sell orders, where u is the common re-quote activity and β_i, σ_i are independent across accounts and sides.

Proposition 4 (netting) *For two desks* $i \neq j$*, the signed flows* $x_i = b_i - s_i = \beta_i - \sigma_i$ *and* x_j *are uncorrelated at every lag, so their contribution to* ρ *in* (3) *is zero; the unsigned channels satisfy* $\mathrm{Cov}(b_i + s_i,\ b_j + s_j) = 4\,\mathrm{Var}(u) > 0$*. For a one-directional ring,* $b_i = u + \beta_i$ *and* $s_i = \sigma_i$*, the signed flows satisfy* $\mathrm{Cov}(x_i, x_j) = \mathrm{Var}(u)$ *and keep their signal, at the price of the extra variance* $\mathrm{Var}(\sigma_i)$ *in* x_i*.*

Proof $\mathrm{Cov}(\beta_i - \sigma_i, \beta_j - \sigma_j) = 0$ by independence; $\mathrm{Cov}(u + \beta_i + u + \sigma_i, u + \beta_j + u + \sigma_j) = \mathrm{Var}(2u) = 4\mathrm{Var}(u)$; and $\mathrm{Cov}(u + \beta_i - \sigma_i, u + \beta_j - \sigma_j) = \mathrm{Var}(u)$. □

Netting therefore removes symmetric two-sided coordination at no cost to the test's validity, and costs power for a ring whose members also submit unrelated orders on the opposite side: the signal-to-noise ratio of the signed flow is lower than that of the buy channel alone by the factor $\mathrm{Var}(\beta_i + u)/\mathrm{Var}(\beta_i + u - \sigma_i)$. Section 5.3 measures both effects. Coordination that is two-sided in a way that does not cancel (for example a fund that buys at one moment in some accounts and sells in others) is not protected by netting.

A blind spot: members that trade against each other.

The same cancellation hides a ring whose members trade against each other. If half of the k members buy and half sell the same quantity at the common moments, which is how cross trading among own accounts looks, the signed flows of the other members cancel in y_{-i} (five of the same sign and six of the opposite sign for $k = 12$), so $\rho \approx 0$ for the signed statistic in either tail, while the unsigned statistic, which correlates buys with buys and sells with sells, keeps the signal. Section 5.3 measures this. Because the unsigned statistic is the one that honest market-making desks defeat, no variant of the test protects against both.

4.5 Deterrence with a fixed inspection capacity

Let a ring pick a strategy θ from a set Θ, earn a random gross gain $g(\theta)$, and be *caught* (event D) with probability $Q(\theta)$ if the regulator inspects the B top-ranked accounts in each of W windows; a caught ring pays $m\, g^+$ where $g^+ = \max(g, 0)$ and m is the fine multiple (5 for individuals and 10 for organisations in Vietnam, Appendix B). Its expected net benefit is

$$U(\theta) = \mathbb{E}[g] - m\,\mathbb{E}[g^+ \mathbf{1}_D]. \tag{5}$$

Proposition 5 (deterrence) *If D is independent of g given θ, then $U(\theta) = \mathbb{E}[g] - mQ\,\mathbb{E}[g^+]$, and $mQ \geq 1$ implies $U(\theta) \leq 0$. If in addition $g \geq 0$, then $U(\theta) \leq 0$ if and only if $Q \geq 1/m$. In the screening model, with (4) and members and windows treated as independent, $Q = 1 - \big(1 - [1 - (1 - q_B)^K]\big)^W$ and the smallest capacity that deters a ring of K accounts over W windows at fine multiple m is*

$$B^* = N\,\Phi\big(\Phi^{-1}(q^*) - S\big), \qquad q^* = 1 - \big(1 - r^*\big)^{1/K}, \quad r^* = 1 - \big(1 - 1/m\big)^{1/W}, \tag{6}$$

whereas random inspection needs $B_{\mathrm{rand}} = Nq^*$. The saving $B_{\mathrm{rand}}/B^* = q^*/\Phi(\Phi^{-1}(q^*) - S)$ is at least 1, increases in S, and tends to 1 as $N \to \infty$ because $S \propto N^{-1/2}$.

Proof $U = \mathbb{E}[g] - mQ\,\mathbb{E}[g^+] \leq \mathbb{E}[g^+](1 - mQ) \leq 0$ when $mQ \geq 1$, since $\mathbb{E}[g] \leq \mathbb{E}[g^+]$. If $g \geq 0$ then $U = \mathbb{E}[g](1 - mQ)$. The expression for Q is the union over members and windows. Solving $Q = 1/m$ for q_B and inverting (4) gives (6); the saving is the ratio of the two capacities, and $\Phi(z - S) \leq \Phi(z)$ for $S \geq 0$. □

Proposition 5 assumes that the detection event is independent of the gain given the strategy. This fails when larger campaigns are both more profitable and more detectable; in the single-market experiment of Appendix C the reported net benefit at $m = 5$ implies that the detected episodes carried about three times the average positive gain, which makes mQ smaller than the independence formula gives and the capacity needed larger, so the capacity results below are optimistic for the regulator. Two consequences are used in Section 6.6. First, a ring that is exposed in many windows by many accounts needs little capacity to be deterred, which is why adaptive rings shrink: they reduce K and the number of active windows. Second, the advantage of screening over random inspection disappears in large markets unless the signal S is boosted, for example by longer windows or better statistics; with $N = 2000$ accounts and a ring of three the saving is about a factor of two.

4.6 A rule that changes the payoff: the block reference price

In the simulator a ring holds a block before the campaign and sells it off-book at the end of each push at a *reference price*, and the exchange can set the reference to the mean mid-price over the last L steps ($L = 1$ is the last price). The reference enters only the payoff, not the dynamics. Write P_t for the mid-price and $\Delta P_t = P_t - P_{t-1}$.

Proposition 6 (block reference price) *With $m_L(t) = \frac{1}{L}\sum_{s=0}^{L-1} P_{t-s}$, the price displacement credited to a block sold at the end t of a push that began at mid-price P_0 is $D_L = D_1 - \sum_{u=0}^{L-2}(1 - \frac{u+1}{L})\Delta P_{t-u}$, where $D_1 = P_t - P_0$: only recent price increments count, discounted linearly with age. If the price rises linearly by D_1 over the last a steps before the sale and is flat before, then*

$$\frac{D_L}{D_1} = \begin{cases} \dfrac{a}{2L}, & L \geq a, \\ 1 - \dfrac{L}{2a}, & L \leq a. \end{cases} \tag{7}$$

If the ring instead holds the price at the top for h further steps, $D_L/D_1 = (a/2 + h)/L$ for $L \geq a + h$, so keeping the full gain requires holding the price for about $L - a/2$ steps.

Proof $P_t - m_L(t) = \frac{1}{L}\sum_{s=0}^{L-1}(P_t - P_{t-s}) = \frac{1}{L}\sum_s \sum_{u<s} \Delta P_{t-u} = \sum_{u=0}^{L-2} \frac{L-1-u}{L}\Delta P_{t-u}$ and $D_L = (m_L(t) - P_0) = D_1 - (P_t - m_L(t))$. For the ramp, the average of the last L prices above P_0 is $D_1 a/(2L)$ when $L \geq a$ (the ramp has mean $D_1/2$ over a of the L steps) and $D_1(1 - L/(2a))$ when $L \leq a$; holding adds h steps at D_1. □

The rule therefore replaces the price level by a linearly decaying memory of the path: a one-day reference ($L = 240$) pays a ring that pushes for $a \approx 60$ steps an eighth of what the last price pays by the ramp law (about 16% for the measured pushes of 72 to 83 steps), and the ring can recover the gain only by holding the price, which is the manipulation the rule penalises. Because the rule changes the payoff only, its effect on a ring that does not adapt can be computed exactly from the same simulated episodes for every L, and compared with the best response; this is done in Section 6.5.

4.7 Evaluating a rule by the ring's best response

Let $\pi \in \Pi$ be a policy (band, block reference L, inspection capacity B, order-level rules), φ a market in a family $\mathcal{M}$, and $U(\theta; \pi, \varphi)$ the ring's net benefit (5). The ring's value is $V(\pi; \varphi) = \sup_{\theta \in \Theta} U(\theta; \pi, \varphi)$. The regulator compares policies by the distribution of $V(\pi; \varphi)$ over $\varphi \in \mathcal{M}$ together with the cost to honest participants $C(\pi; \varphi)$ (spread, depth, volatility); it deems π *deterring* in φ if $V(\pi; \varphi) \leq 0$.

Proposition 7 (best response on a finite strategy set) *Let $\hat{\Theta} \subset \Theta$ be finite, let $\hat{\theta}$ maximise the mean over n selection episodes, and let $\hat{U}$ be the mean over n further, independent episodes. Then $\mathbb{E}[\hat{U} \mid \hat{\theta}] = U(\hat{\theta}; \pi, \varphi) \leq V(\pi; \varphi)$. Hence the reported ring benefit is unbiased for the benefit of the chosen strategy and a lower bound in expectation of the true best response, and a policy that appears deterring on $\hat{\Theta}$ may not be deterring against a larger set.*

Proof $\hat{U}$ is independent of $\hat{\theta}$ given the market, so conditioning removes the selection bias; and $U(\hat{\theta}) \leq \sup_{\Theta} U$. □

The proposition is the reason why the reported ring benefits are lower bounds on the ring's true best-response value: with a better search the ring would do at least as well. A policy that appears deterring on the menu ($\hat{U} \leq 0$) need not be deterring ($V \leq 0$), so the deterrence results and the reductions of the ring's gain by a rule are optimistic for the regulator, and are to be read as upper bounds on what a regulator can claim. The family $\mathcal{M}$ is drawn by approximate Bayesian rejection: parameters of the honest market are sampled, honest-only episodes are simulated, and a draw is kept if its cancel share, retail share and daily volatility are within tolerance of the real values (Section 3.3). A conclusion that holds for every accepted market is more credible than one tuned to a single calibration, but the tolerance box is wide, and markets that match the moments need not match anything else.

5 Experiments I: the screening statistic

We check the statements of Section 4 on the test in three steps (the deterrence and block-price laws are checked in Section 6): with synthetic flows where the truth is known (validity, budget, power law, netting), in the agent-based market with a ring and honest groups (sensitivity, robustness, comparison with other detectors), and on real account-level order flow (Section 7.1). Unless stated, the agent-based results are for the default simulator (100 to 170 accounts in a window), one ring per episode, episodes of 6000 steps (19 windows of 300 steps), and the metric is *catch@B*: the share of ring-active windows in which one of the B top-ranked accounts is a ring member, with 95% bootstrap intervals over 24 test episodes.

5.1 Validity and the inspection budget (Propositions 1 and 2)

Figure 3a gives the rejection rate of the implemented signed test against its nominal level for exchangeable flows. For independent Gaussian flows and for Poisson counts the rate is at the nominal level (0.0100 and 0.0102 at the 1% level, 0.051 and 0.052 at

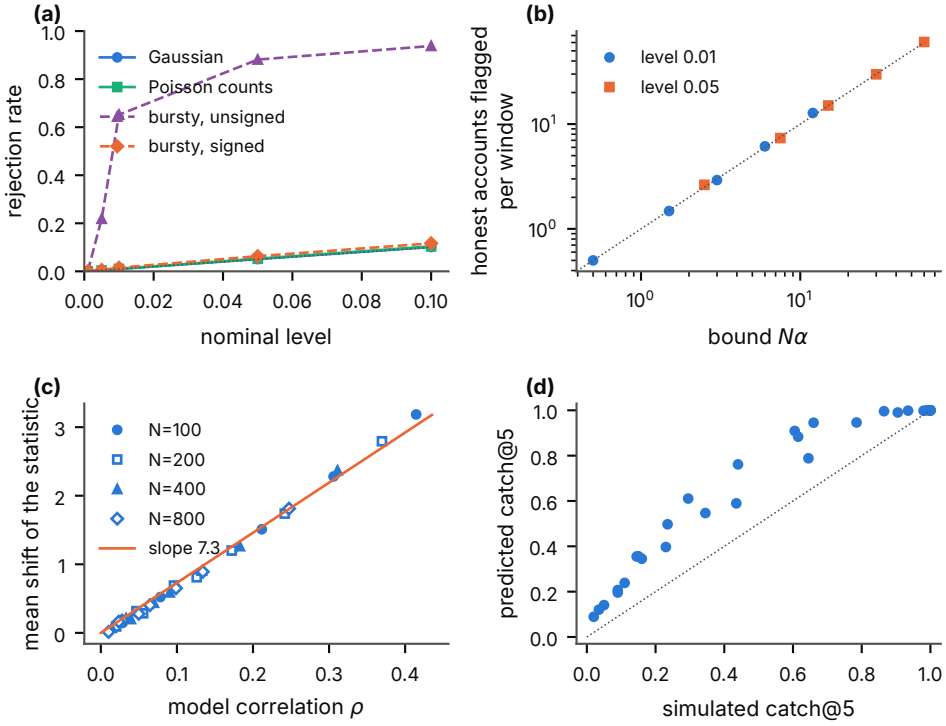


Fig. 3 Monte Carlo checks of the propositions with the implemented test. (a) Rejection rate against nominal level for exchangeable flows (Gaussian, Poisson) and for flows with a shared bursty regime (signed and unsigned variants); dotted line: nominal. (b) Honest accounts flagged per window against the bound $N\alpha$ at levels 1% and 5%. (c) Mean shift of a ring member's statistic against the model correlation ρ of Proposition 3, for N from 100 to 800, k from 4 to 32 and $\eta \in \{0.1, 0.3\}$; line: fitted slope. (d) Predicted against simulated catch@5 for the same 32 settings

5%, 18,000 tests each). Rejection is zero at the 0.1% level because $p \geq 1/T = 1/300$ (Proposition 2b). The condition of the proposition is violated by flows with a shared regime, bursts of activity that all accounts follow: the unsigned test then rejects 65% of honest accounts at the 1% level, and the signed test, in which symmetric bursts cancel, 1.5%. Figure 3b confirms the expected number of honest flags per window against the bound $N\alpha$ for N from 50 to 1200, and the Bonferroni rule at level 5% flags nothing for any of these N, as the resolution limit says. We therefore use the test as a ranking device and report observed rates.

5.2 The power law (Proposition 3)

In a Gaussian-flow market with N from 100 to 800 accounts, ring sizes from 4 to 32 and signal-to-noise ratios $\eta \in \{0.1, 0.3\}$ (32 settings, 200 windows each), the mean shift of a member's statistic over the honest mean is linear in the model correlation ρ of (3) (Figure 3c; through the origin, $R^2 = 0.993$; fitted slope $\sqrt{T_{\text{eff}}} = 7.3$, that is $T_{\text{eff}} \approx 53$ of $T = 300$, within a factor of two of $T/(2\tau + 1) = 100$ for the smallest tolerance).

The predicted catch@5 from (4), with members treated as independent, tracks the simulated one (correlation 0.96) but overstates it by 0.13 on average (Figure 3d), because members' statistics are positively dependent through the shared signal and the shared aggregate. The law is therefore a good guide to *how* detection varies with N and k and a poor guide to its level at small k.

In the agent-based market the same law explains the dependence on the number of accounts. The ring there has structure the Gaussian model omits: only the m_{push} members that push together are co-active, so the number of co-active accounts does not change with the ring size K, and the signal is proportional to the push probability. The model fitted to the three one-factor sweeps is

$$\text{shift} = s_0 \, \frac{p_{\text{push}}}{0.8} \left(\frac{100}{N} \right)^{1/2}, \qquad \text{catch@}B = a \left[1 - (1 - q_B)^K \right], \tag{8}$$

with q_B from (4), two constants ($s_0 = 1.20$ standard units of the statistic at $N = 100$, and a ceiling $a = 0.90$ for ring-active windows without signal) and the exponent $1/2$ taken from the theory. The 11 points are fitted with a mean absolute error of 0.043 (Figure 4); holding one sweep out and predicting it from the other two gives errors of 0.045 for ring size, 0.085 for the number of accounts (where the $N^{-1/2}$ law is a pure prediction) and 0.088 for the push probability. When the exponent is left free it is 0.32, below the theoretical 0.5; with 11 points we cannot say whether this reflects the non-Gaussian flows or noise, but both give detection falling steeply with N: catch@5 is 0.91 at 100 accounts and 0.28 at 708.

5.3 Honest groups and netting (Proposition 4)

With synthetic flows, six market-making-like desks that re-quote on both sides at the same moments are flagged at the 1% level in 80% of cases by the unsigned test and in 1.1% by the signed test, whereas a one-directional ring is flagged in 37% by the unsigned and 17% by the signed test: netting removes the desks and costs power for the ring, as the proposition says. In the agent-based market, Table 5 and Figure 5 give catch@5 with and without honest groups. The baselines that need no labels fail on the ring: the order-to-trade ratio is dominated by market makers, which sit far above manipulators, and an isolation forest on account features finds almost nothing. The unsigned test finds the ring in a clean market but collapses when market-making desks are present, because the desks are flagged at 71% (Table 6) and fill the top of the ranking. Netting buys against sells removes the effect of the desks (their flag rate falls to 0 in that scenario and catch@5 stays at 0.88). A fund that splits orders across sub-accounts is coordinated in time without being a manipulator, and is flagged at 6% whatever the variant; it reduces catch@5 to 0.74. With all three honest groups together catch@5 is 0.52 for the signed test and 0.23 for the unsigned one, and the desks are flagged at 18% even after netting. A gradient-boosting model trained with labels on rings in the clean market stays at 0.98–1.00 in every scenario.

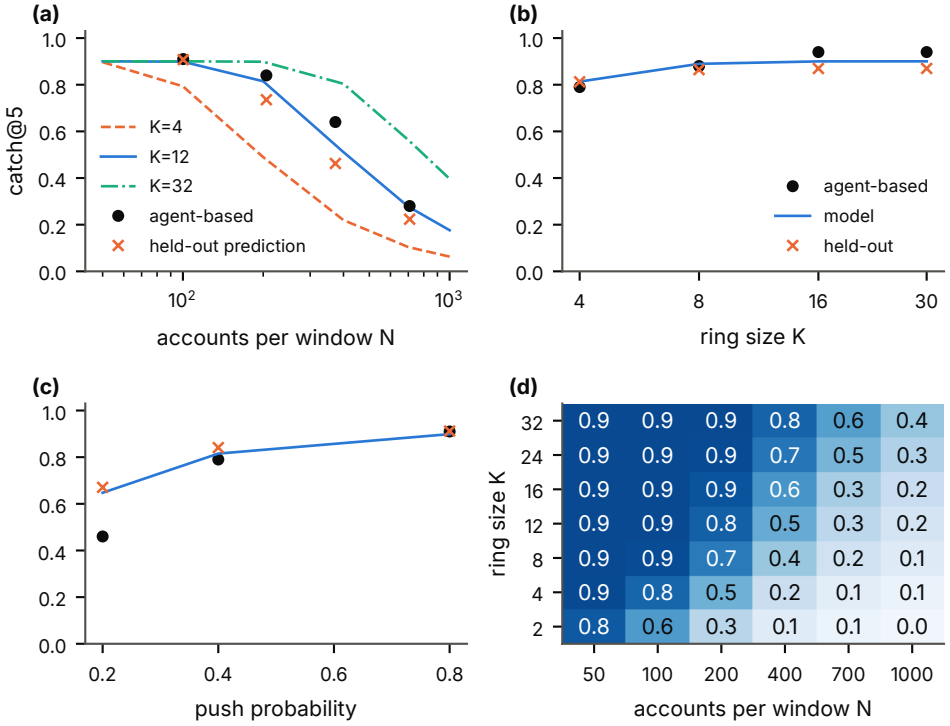


Fig. 4 The scaling law against the agent-based market. (a)–(c) catch@5 against the number of accounts, the ring size and the push probability: points are simulated, lines are the model (8) fitted to all sweeps, crosses are predictions with the plotted sweep held out. (d) Model catch@5 over the number of accounts and the ring size at push probability 0.8

detector	clean	+MM desks	+fund splitting orders	+bot fleet	all three
order-to-trade ratio	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.13 [0.05, 0.22]	0.09 [0.04, 0.14]
isolation forest (no labels)	0.01 [0.00, 0.02]	0.01 [0.00, 0.02]	0.00 [0.00, 0.00]	0.01 [0.00, 0.03]	0.02 [0.00, 0.03]
coincidence test, unsigned (no labels)	0.85 [0.78, 0.91]	0.32 [0.20, 0.46]	0.71 [0.65, 0.77]	0.86 [0.80, 0.91]	0.23 [0.15, 0.31]
coincidence test, signed (no labels)	0.87 [0.81, 0.92]	0.88 [0.82, 0.92]	0.74 [0.66, 0.80]	0.85 [0.79, 0.90]	0.52 [0.42, 0.63]
gradient boosting (labels, clean market)	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0.99 [0.98, 1.00]	0.98 [0.96, 1.00]	0.98 [0.97, 1.00]

Table 5 catch@5 in a clean market and with honest groups acting in step (95% bootstrap intervals).

Cross trading.

The netting that protects the signed test against market-making desks hides a ring whose members trade against each other (Section 4.4). With synthetic count flows (300 accounts, a ring of 12, 200 windows), a one-directional ring is in the top five in 99% of the windows for the signed test and 99% for the unsigned test; a ring of six buyers and six sellers at the common moments is found in 14% (positive tail) and 25% (negative tail) of the windows by the signed test and in 85% by the unsigned test. In the agent-based market, where cross trades are a share of the orders of an otherwise one-directional ring, the share does not matter: catch@5 is 0.91, 0.89, 0.94 and 0.92

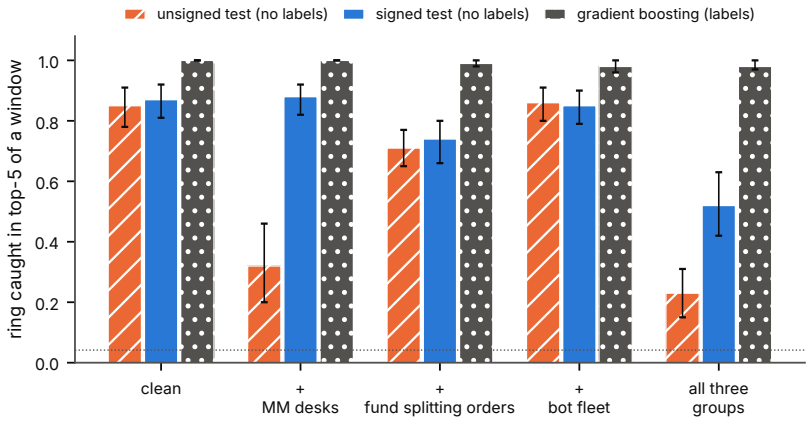


Fig. 5 Share of ring-active windows in which a ring member is among the five top-ranked accounts, with 95% bootstrap intervals over 24 episodes. The dotted line is the chance level. The unsigned test is crowded out by market-making desks; the signed test is not. Gradient boosting is trained with labels on rings in the clean market

scenario	honest group	members scored	flagged, unsigned	flagged, signed
+ market-making desks	market-making desks	2622	0.711	0.000
+ fund splitting orders	fund sub-accounts	5277	0.061	0.063
+ bot fleet	bot fleet	3571	0.013	0.016
all three	market-making desks	2698	0.665	0.182
all three	fund sub-accounts	5202	0.050	0.053
all three	bot fleet	3732	0.029	0.037

Table 6 Honest group members flagged at $p \leq 0.01$ (nominal 1%).

for cross shares of 0, 0.2, 0.35 and 0.5 (24 episodes each). Four of the ten documented cases are described as cross trading among own accounts (Appendix A); against a purely cross-trading ring the signed test is blind, and the unsigned test, which is not, is the one that honest desks crowd out.

5.4 Comparison with other detectors

Figure 6 gives receiver-operating and precision–recall curves of the account-window scores for the unsigned and signed tests, the supervised model and an isolation forest, in the clean market, with market-making desks and with all three honest groups. The signed test has an area under the curve of 0.85, 0.87 and 0.73 in the three scenarios and the unsigned test 0.84, 0.79 and 0.72; the supervised model 0.99, 0.99 and 0.98; the isolation forest and the order-to-trade ratio are below chance (0.36 to 0.46): the order-to-trade ratio is dominated by market makers, and the most unusual accounts by their features are not the ring members. The average precision of the label-free tests is low, 0.24, 0.27 and 0.075 for the signed test (0.22, 0.10 and 0.06 unsigned) against 0.89 to 0.94 for the supervised model and a share of 3% to 4% of ring members among the account-windows. The label-free tests are informative over the whole ranking but

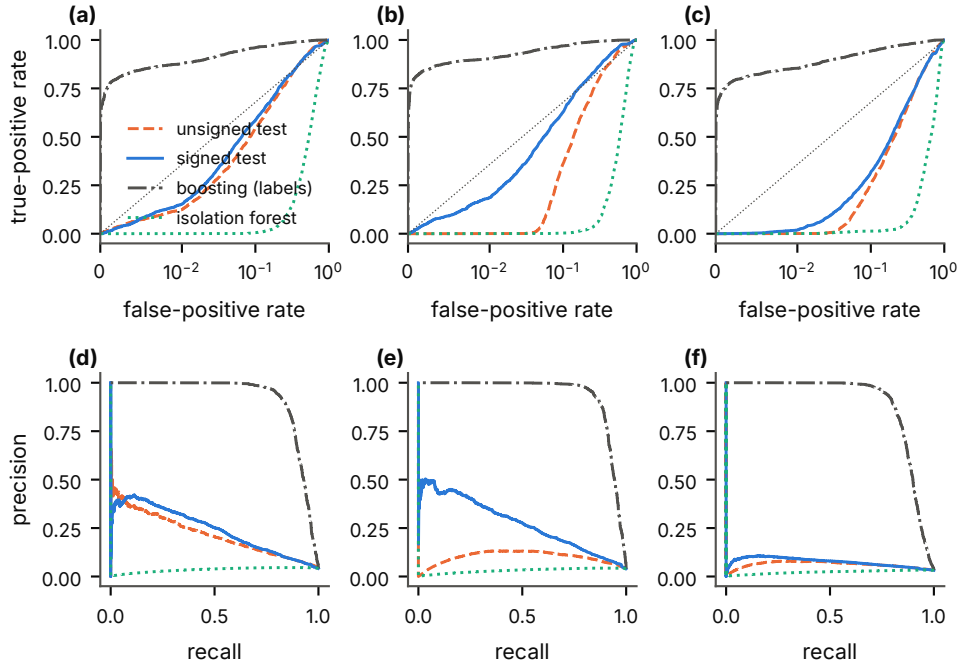


Fig. 6 Receiver-operating (top) and precision–recall (bottom) curves of account-window scores for ring membership, in a clean market (a, d), with market-making desks (b, e) and with all three honest groups (c, f); 24 test episodes each. The supervised model is trained with labels in the clean market only

their precision is high only at its very top, which is what catch@5 measures; they are not a classifier. The netting in the signed test matters most with market-making desks (average precision 0.27 against 0.10).

5.5 Sensitivity

Figure 7 and the ablation table of Appendix D vary the design. The tolerance set and the window length matter little for catch@5 (between 0.88 and 0.91), but catch@1 rises from 0.41 to 0.58 as the window grows from 150 to 600 steps, as the dependence of the shift on T_{eff} in Proposition 3 suggests. Sensitivity falls with the number of accounts and rises with ring size (0.79 for four accounts, 0.94 for 16 or more) and with the ring's activity (0.46, 0.79, 0.91 for push probabilities 0.2, 0.4, 0.8), as the law predicts (Figure 4). The dependence on the number of accounts is the main weakness of ranking by top-B; a rule based on finer resampling and a false-discovery-rate bound would scale differently, and we have not tested it.

What the label-free test is for.

The model trained with labels is better in every simulated scenario. The label-free test is useful when there are no labels, which is the regulator's position, and when one

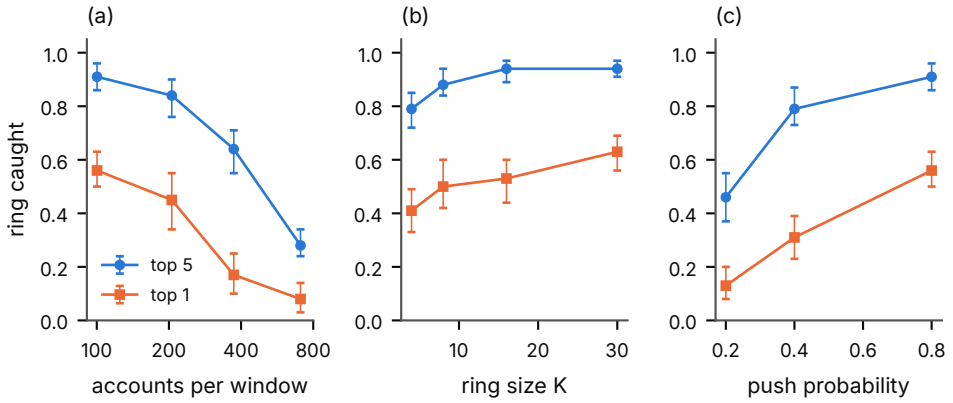


Fig. 7 Sensitivity of the signed test in a clean market: (a) number of accounts per window, (b) ring size, (c) push probability of the ring. Bars are 95% bootstrap intervals over 16 episodes

cannot trust that a model trained on simulated rings transfers to real ones. Section 7.1 shows that the second concern is justified for the test itself: on real order flow it also faces a background the simulator does not contain.

5.6 Transfer across simulated markets

A supervised model is trained on labelled rings, and the regulator has none, so the natural question is whether a model trained in one simulated configuration works in another. We train the boosting model on 24 labelled episodes of the generic simulator configuration (default ring) and apply it, unchanged, to ten markets of the Vietnam preset: the default preset and three accepted markets from each of the index-level, stock-level and trend-free families of Section 6, with 16 test episodes per market. Table 7 compares it with the label-free test and with the same model trained on labelled episodes of the test market itself. The transferred model loses little against the in-market model (catch@3 of 0.86–0.94 against 0.95–0.97) and is far better than the label-free test, which finds a ring of 12 accounts in 28–38% of ring-active windows at three inspections in the three families with fewer accounts and in 1% in the trend-free family. These markets have three to five times as many fundamental traders (118–186 against 34–46) and more market-making activity, so the signal index of Proposition 3 is much smaller, which is consistent with the law of Section 5.2 but was not fitted to these markets.

The result limits what can be concluded in two ways. First, all these markets and the rings in them are produced by the same simulator, so features that identify a ring (such as the order-to-trade ratio and the timing of its orders) are distinctive in every one of them; transfer across simulator parameters says nothing about transfer to real rings, whose behaviour we do not know. Second, the label-free test is not a substitute for a supervised model in a large market: it is the only one of the two whose false-alarm rate can be stated exactly, and in our trend-free markets that guarantee comes with almost no power against a ring of this size.

market family (markets)	detector	catch@1	catch@3	catch@5	catch@10
default preset (1)	label-free test	0.15	0.28	0.37	0.50
	boosting, trained on the generic simulator	0.75	0.90	0.92	0.94
	boosting, trained in this market	0.93	0.95	0.95	0.96
index level (3)	label-free test	0.20	0.38	0.46	0.58
	boosting, trained on the generic simulator	0.76	0.90	0.92	0.94
	boosting, trained in this market	0.96	0.97	0.97	0.98
stock level (3)	label-free test	0.19	0.36	0.46	0.58
	boosting, trained on the generic simulator	0.85	0.94	0.95	0.96
	boosting, trained in this market	0.96	0.97	0.98	0.98
trend-free (3)	label-free test	0.00	0.01	0.02	0.03
	boosting, trained on the generic simulator	0.73	0.86	0.89	0.93
	boosting, trained in this market	0.94	0.96	0.97	0.98

Table 7 Share of ring-active windows in which a ring member is among the B top-ranked accounts (mean over markets, 16 test episodes per market, default ring): the label-free test, a boosting model trained on the generic simulator and applied unchanged, and the same model trained on 24 labelled episodes of the test market

6 Experiments II: rules against a ring that adapts

6.1 Design

Levers.

The daily price band (7% today, 10%, 15% or none); the reference price of off-book block trades (the last price, or the mean over the last $L = 60$ or 240 steps); inspection of the B top-ranked accounts per window by the signed test, with a fine of m times the illegal gain; a minimum resting time before an order may be cancelled (5 or 15 steps); a fee per cancelled order; a larger tick; a circuit breaker. With $m = 5$ and $m = 10$, deterrence needs a probability of detection above $1/m$ (Proposition 5).

Manipulators.

The ring chooses a strategy from the 11-parameter space of Table 2. Because single evolution-strategy runs were noisy and non-monotone across policies, every policy faces the same set of four ring strategies (the strongest found under different policies in one market, the default ring, and the strongest unconstrained one); the ring picks the best on 12 selection episodes and the tables report 12 separate episodes (Proposition 7: a lower bound on a fully adaptive ring). The one unrestricted evolution-strategy search we ran, in a single market, gave non-monotone gains across policies (101k with no band, 49k with a 7% band, 94k with 10%, 84k with 15%; Appendix C), so the noise of such a search is of the order of the effects we interpret, which is why the menu is cross-evaluated; we have not shown that the menu is near the optimum for each policy, and every conclusion below is conditional on it (Section 6.4 adds 24 random strategies to the menu in all 16 markets). The spoofer, the pump-and-dump agent and the wash pair use fixed strategies and are evaluated for what they achieve; for them the effect of a rule is an upper bound.

policy	median gain	range over markets	median ratio to no policy	median detection prob.	deterred at $m = 5$
no price band	110.0k	86.7k to 162.1k	1.00	–	–
band 7% (HOSE today)	105.0k	89.2k to 164.6k	0.97	–	–
band 7% + block at 60-step mean	38.4k	11.7k to 90.8k	0.35	–	–
band 7% + block at 240-step mean	15.3k	-0.9k to 23.5k	0.11	–	–
band 7% + inspect 1 per window	16.8k	4.5k to 51.6k	0.17	17%	2/6
band 7% + inspect 3 per window	16.8k	4.5k to 51.6k	0.18	42%	5/6

Table 8 Index-level family of six calibrated markets: the best-responding ring's gain (tick-lots) against each policy, the median probability that a member is inspected in some window of an episode, and the number of markets in which the ring's expected net benefit is non-positive at a fine of five times the gain.

Families of markets.

Market parameters (numbers of noise and fundamental traders, market-maker activity and sensitivity to displayed depth, volatility of the latent value, momentum activity) are drawn at random, honest-only markets are simulated, and a draw is kept if its cancel share (0.27–0.37), retail share (0.66–0.85) and daily volatility (0.9–1.4% for the index-level family, 1.8–2.8% for the stock-level family) are within tolerance of the real ones, an approximate Bayesian rejection. We keep six markets per family (6 of 27 draws and 6 of 62 draws) and report the distribution of each outcome over markets. Acceptance used only two episodes per draw, so it is noisy: re-measured over 16 episodes (Appendix D), the volatility is inside the tolerance in five of six index-level and four of six stock-level markets, the cancel share in five of six in each family and the retail share in all, and the excursions are of up to about two standard errors of the re-measurement for the volatility and of at most 0.008 for the cancel share. Figure 2 compares the first accepted markets with real returns.

6.2 The ring against each rule

Figure 8 and Table 8 give the ring's gain in the index-level family. The daily price band does not change the ring's gain (median ratio to no policy 0.97; the ring moves the price by about 0.5%, far inside a band of 7%), and widening it to 10% or 15%, which the securities commission is studying, changes nothing in the single market where we tried it (Appendix C). Pricing blocks at the trailing mean cuts the gain, Section 6.5 analyses how much. Inspection deters the ring only with enough capacity: with three inspected accounts per window (0.5% of accounts) the ring's expected net benefit at $m = 5$ is non-positive in five of six markets, with one inspected account in two of six; the median probability of being caught is 42% and 17%.

Stock-level family.

The case stocks are more volatile than the index, so we repeat the exercise with six markets whose daily volatility is 1.8–2.8% (Table 9; same ring strategy set, not re-optimised). The ring's gain with no policy is larger (median 164k against 110k at index level). The conclusions on the band and on inspection are unchanged in sign: the band leaves the gain at a median 99% of its value. Under inspection the best-responding ring shrinks its operation: its median gain falls to 3% of the no-policy gain (6k), with a median probability of being caught of 8% with one inspected account per window and 25% with three. The ring's expected net benefit at $m = 5$ is non-positive in three

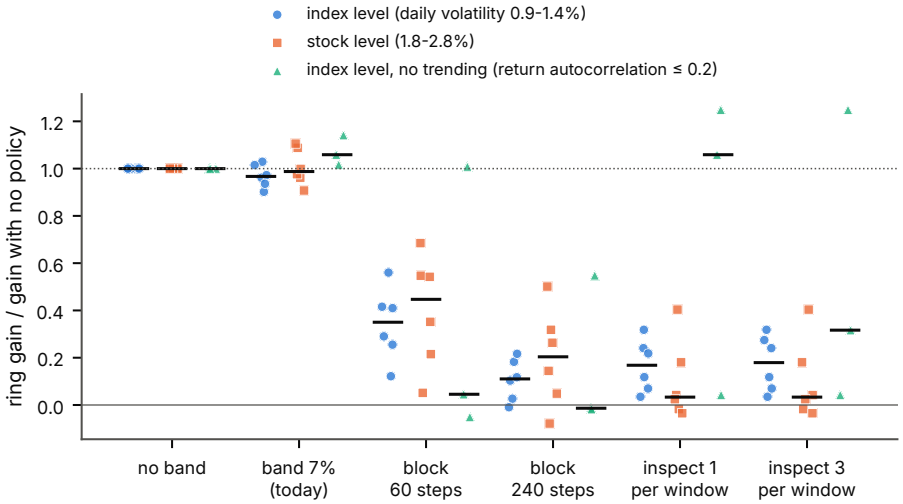


Fig. 8 The best-responding ring's gain relative to its gain with no policy, one point per calibrated market and the median as a line. Daily band 7% in all policies but the first

policy	median gain	range over markets	median ratio to no policy	median detection prob.	deterred at $m = 5$
no price band	164.4k	101.2k to 261.4k	1.00	–	–
band 7% (HOSE today)	154.9k	111.9k to 261.1k	0.99	–	–
band 7% + block at 60-step mean	66.5k	8.4k to 179.0k	0.45	–	–
band 7% + block at 240-step mean	33.7k	-7.9k to 131.0k	0.20	–	–
band 7% + inspect 1 per window	6.3k	-9.1k to 40.8k	0.03	8%	3/6
band 7% + inspect 3 per window	6.3k	-9.1k to 40.8k	0.03	25%	5/6

Table 9 Stock-level family of six calibrated markets (daily volatility 1.8–2.8%): the ring's gain against each policy, as in Table 8.

of six markets with one inspected account and in five of six with three; in the three markets where one inspected account does not deter, the ring still nets up to 41k, and in the one market where three do not, 7k. The detection probabilities are lower than at index level, yet deterrence is not worse, because the ring's remaining gain is smaller relative to the fine; we have not separated the two effects. In both families the price band has no effect, block pricing and inspection have large ones, and no single rule removes the ring's gain in every market.

6.3 Markets without trending

The simulated daily returns of the families above trend (lag-1 autocorrelation 0.4 to 0.6, Figure 2c), whereas the daily returns of VN30 and of HOSE stocks have no serial correlation. The trending comes from the slow adjustment of the price to the latent value, and a ring may profit from it. To check whether the conclusions depend on it, we drew a third family with many more fundamental traders (80 to 220 against 25 to 60, so that the price follows the latent value within a day), more noise traders and market makers, and accepted a draw only if, besides the volatility (0.9–1.5%) and

policy	median gain	range over markets	median ratio to no policy	median detection prob.	deterred at $m = 5$
no price band	43.4k	-40.2k to 63.9k	1.00	–	–
band 7% (HOSE today)	48.1k	-38.5k to 65.0k	1.06	–	–
band 7% + block at 60-step mean	0.6k	-58.4k to 52.2k	0.05	–	–
band 7% + block at 240-step mean	-0.8k	-52.0k to 28.4k	-0.01	–	–
band 7% + inspect 1 per window	19.9k	-19.1k to 64.6k	1.06	4%	2/4
band 7% + inspect 3 per window	6.9k	-19.1k to 64.6k	0.32	4%	2/4

Table 10 Family of four markets without trending daily returns (lag-1 autocorrelation of at most 0.2): the ring's gain against each policy, as in Table 8.

the cancel share, the lag-1 autocorrelation of daily returns of four honest episodes of 6000 steps was at most 0.2 in absolute value; the retail-share tolerance was widened to 0.55–0.85 because more fundamental traders lower it. Four of 29 draws were accepted (autocorrelation 0.03 to 0.20, volatility 1.05–1.37%, retail share 0.62–0.65), so this family has four markets, and its moments were not re-measured.

Table 10 gives the results. The band again has no effect (median ratio 1.06). In three of the four markets the ring is profitable without a rule (35k to 64k); in the fourth it loses money (−40k) whatever the rule. Pricing blocks at the trailing mean removes the gain in two of the three profitable markets (to −0.5k and −1.1k at 240 steps) and in the third leaves 55% of it (28k against 52k), where the ring's gain comes mostly from trading. Inspection is less effective than in the other families: the median probability of being caught is 4% with one and with three inspected accounts, and the ring's expected net benefit at $m = 5$ is non-positive in two of four markets, one of which is the market where the ring loses money anyway; in the market where it is not caught at all it nets 65k. Markets with more accounts per window (about 630 to 1070 noise and fundamental traders against 400 to 670 before) are the markets where the power law predicts weaker detection (Section 5.2). The conclusion on the band and on the block price therefore survives the removal of the trending; the conclusion on inspection depends on the number of accounts, and the capacity needed grows with it.

6.4 Is the four-strategy menu enough?

The review of any best-response result starts with the strategy set. In each of the 16 accepted markets we gave the ring, for three policies (the band alone, the band with block prices at the 240-step mean, and the band with inspection of three accounts per window), a menu of 28 strategies: the four of the ensembles and 24 drawn uniformly in the 11-parameter space of Table 2. Selection has two stages on separate episodes (all 28 on six episodes; the four menu strategies and the five best on twelve more), and the chosen strategy is scored on twelve fresh episodes, so that the maximum over many noisy estimates does not enter the reported gain; a single stage with six episodes picked noise in a trial run (a reported gain of 23k against 93k for the menu of four in one cell). Table 11 compares the ring with the menu of four with the ring with the larger menu.

Against the band alone and against block pricing the larger menu changes nothing: in 15 of 16 markets the strategy chosen from 28 is one of the four, and the medians are identical. The share of the gain that survives block pricing at 240 steps is 0.32 in both cases, and in every market the gain falls. Against inspection the larger menu chooses a random strategy in 13 of 16 markets, but its gain on the fresh episodes is not

		gain, band only (thousand)		block at 240 steps: ratio to band only		inspection of 3: markets with net benefit > 0	
family	markets	menu of 4	larger menu	menu of 4	larger menu	menu of 4	larger menu
index level	6	124	124	0.25	0.25	1/6	1/6
stock level	6	224	224	0.41	0.41	0/6	0/6
trend-free	4	34	30	0.37	0.23	2/4	3/4
all markets	16	106	106	0.32	0.32	3/16	4/16

Table 11 The ring's best response with the menu of four strategies and with a menu of 28 (the four and 24 random ones): median gain against the band alone, ratio of the gain under block pricing at 240 steps to the gain with the band alone, and number of markets in which a ring facing inspection of three accounts per window keeps a positive net benefit at fine multiple 5. Two-stage selection and 12 fresh reporting episodes per cell

higher than the menu's (median difference −10k over these 13 cells, higher in 5), and the number of markets in which inspection of three accounts leaves the ring a positive net benefit at $m = 5$ goes from 3 to 4 of 16, the extra market being in the trend-free family. We read this as no evidence that the four-strategy menu understates the ring's best response by much in the three policies, with two caveats: 24 random points in 11 dimensions are a weak search, and an adaptive search against inspection would exploit the structure of the detector better than random draws do. The deterrence conclusions therefore stay upper bounds for the regulator (Proposition 7), by a margin that this check does not bound from above.

Intervals over markets.

The unit of replication of the ensemble results is the market, and with six markets per family the intervals are wide. Over the 16 markets of the ensembles, the ring's gain with block prices at the 240-step mean falls in all 15 markets where the band-only gain is positive (exact sign test $p < 0.001$); the median ratio to the band-only gain is 0.12 (bootstrap 95% interval over markets −0.01 to 0.24; index level 0.11, stock level 0.20, trend-free −0.01), and at 60 steps 0.36 (0.13 to 0.55). Inspection of three accounts per window leaves the ring a positive net benefit at $m = 5$ in 4 of 16 markets (Wilson 95% interval 10 to 49%), with a median probability of detection of 21%. The fresh episodes of this section give a larger median ratio for block pricing (0.32), because the ratio of two noisy gains with a small denominator is itself noisy, but not a different sign. We therefore state the direction of the block-price and band results with confidence and their size as a range.

6.5 The block reference price: law, ex-post evaluation and best response

Because the reference price enters only the ring's payoff (Section 4.6), the effect of the rule for every L can be computed exactly from the same simulated episodes. For each market of both families and each of the four strategies we simulate 12 selection and 12 reporting episodes once and score them with $L \in \{1, 5, 15, 30, 60, 120, 240, 480\}$. Table 12 and Figure 9 report the median over markets. The *block term* is the value of the block alone (block size times credited displacement); the total gain adds trading profit, which the rule does not touch. The rule removes most of the block term: a reference of 60 steps leaves 42% (index level) and 39% (stock level) of it, a one-day

	index level				stock level			
L	block term	law	total, static	total, adaptive	block term	law	total, static	total, adaptive
15	0.87	0.90	0.88	0.88	0.84	0.90	0.91	0.91
30	0.72	0.81	0.76	0.76	0.67	0.81	0.83	0.83
60	0.42	0.61	0.52	0.52	0.39	0.61	0.68	0.64
120	0.22	0.32	0.36	0.45	0.21	0.32	0.59	0.48
240	0.13	0.16	0.28	0.28	0.02	0.16	0.44	0.44
480	-0.15	0.08	-0.03	0.09	-0.41	0.08	0.12	-0.01

Table 12 Block-price rule, median over six markets per family: share of the block term and of the total gain retained when the block is priced at the mean of the last L steps instead of the last price, and the prediction of the linear-ramp law.

reference 13% and 2%. The linear-ramp law of Proposition 6, with the measured push duration $a = 72$ to 83 steps, has the right shape and order of magnitude but overstates the retained value at $L \geq 30$ (0.61 against 0.42 at $L = 60$ at index level, 0.16 against 0.13 at $L = 240$), because the real price path is not a ramp ending at the sale. The total gain falls to a median 52% (60 steps) and 28% (240 steps) of its value at index level and to 68% and 44% at stock level, as the trading profit remains; with a reference of two days ($L = 480$) the median total gain is about zero (-0.03 at index level, 0.12 at stock level), and in markets where the unconstrained gain is small the ring loses money already at 60 to 240 steps. Re-selecting the strategy ("adaptive") changes the median total gain by 0.13 or less in either direction: within the four strategies the ring finds no good counter, and the gap between the adaptive and the static curve is the value of adaptation. The independent episodes of the earlier ensemble run (Table 8) give a lower retained share of the total gain (35% and 11% at index level, 45% and 20% at stock level for 60 and 240 steps); with episode-to-episode standard errors of half the mean in gain or more, we read the evidence as: a one-day reference leaves the ring between roughly one tenth and one half of its gain, and removes most of the block value.

Sensitivity to the block size.

The block of 3000 lots is an assumption of the simulator, and it matters: the rule acts on the block term only, so its effect on the total gain depends on how much of the gain comes from the block. The size enters only the payoff, so a different size can be evaluated exactly on the same episodes (Appendix D). With a block of 10,000 lots the ring keeps 19–21% of its gain at a one-day reference; with the assumed size 28–44%; with a block of 1000 lots 60–64%, and of 300 lots 83–87%, because trading profit then dominates. The rule is effective against a ring whose gain is realised through off-book blocks and has no effect on a ring that profits through trading on the book.

6.6 Inspection capacity and the deterrence frontier

Proposition 5 turns the power law into a capacity requirement. With the constants fitted in Section 5.2 ($s_0 = 1.20$), Table 13 gives the number of accounts that must be inspected per window to deter a ring of three accounts active in four windows, and the saving over random inspection. In a market with 600 active accounts per window,

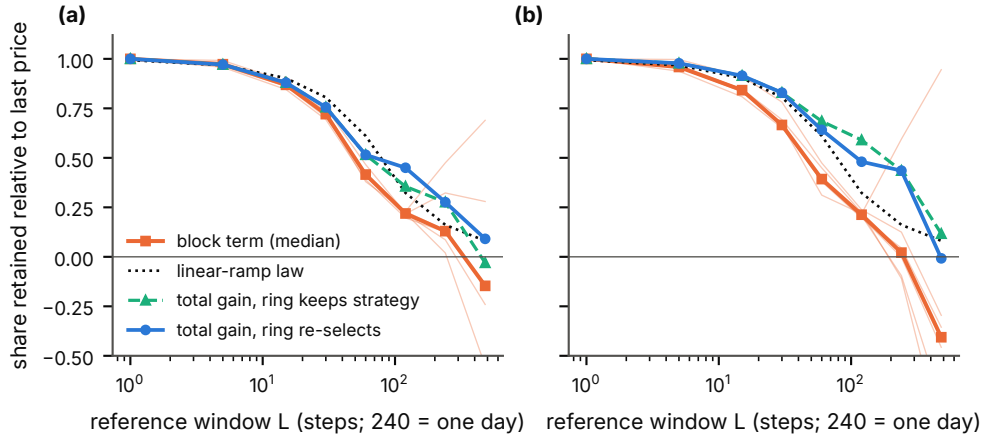


Fig. 9 Share of the ring's block term and of its total gain retained against the reference window L, for the index-level (a) and stock-level (b) families: median over six markets of the block term (thick orange; thin lines are the six markets), the linear-ramp law of Proposition 6, and the total gain for a ring that keeps its strategy and for a ring that re-selects it

	fine 5× gain			fine 10× gain		
N	screening	random	saving	screening	random	saving
100	0.05	1.8	36.5	0.02	0.9	50.2
300	0.81	5.5	6.8	0.32	2.6	8.1
600	2.99	11.1	3.7	1.25	5.2	4.2
1000	6.81	18.4	2.7	2.93	8.7	3.0
2000	18.48	36.8	2.0	8.17	17.5	2.1

Table 13 Inspected accounts per window needed to deter a ring of three accounts active in four windows (Proposition 5 with $s_0 = 1.20$): by screening and by random inspection, and the saving, for fine multiples 5 and 10.

screening deters at a fine of five times the gain with about 3 inspected accounts per window (0.5% of accounts) against 11 for random inspection, and at ten times with 1.2 against 5.2; doubling the fine multiple therefore saves a factor of about 2.4 in capacity, which gives the regulator two exchangeable instruments (Section 8). The advantage of screening shrinks as the market grows (saving of 3.7 at 600 accounts and 2.0 at 2000). The model's detection probability for a ring of three accounts in four windows with one and three inspected accounts (8% and 20%) is close to the probability found for the best-responding ring in the stock-level family (8% and 25%) and below that of the index-level family (17% and 42%; Figure 10a). We do not know which sizes the best-responding ring chose, so this agreement is suggestive, not a test.

6.7 Other manipulation types and the cost to honest investors

Table 14 and Figure 11 report, for the first calibrated market, what a fixed spoofer, pump-and-dump agent, wash pair and ring achieve under order-level rules and what

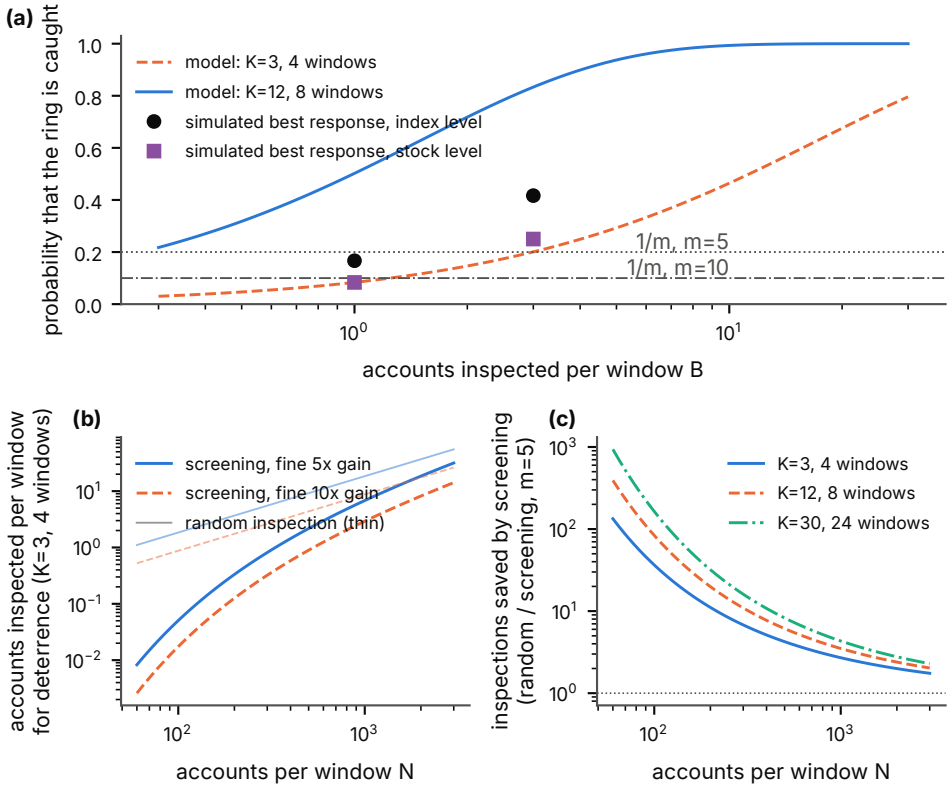


Fig. 10 Deterrence frontier. (a) Probability that the ring is caught against the number of inspected accounts per window at $N = 600$ for two ring sizes (model, solid and dashed), with the thresholds $1/m$ for $m = 5, 10$ (horizontal lines) and the median probability of the best-responding ring in the two simulated families (points). (b) Accounts to inspect per window for deterrence of a ring of three accounts active in four windows against N, by screening (thick) and by random inspection (thin). (c) Inspections saved by screening against N for three ring sizes and activity levels, fine five times the gain

the rules cost honest participants. In this market, minimum resting times and cancel fees do not reduce what the manipulators achieve: spoofing profit is 872 (s.e. 125) with no rule and 1,091 to 1,380 with minimum resting times or fees, and the spoofing price impact is higher, not lower, with a longer minimum resting time (0.26 against 0.11 ticks); pump impact is lower under cancel fees (0.31 against 0.47; 1.3 to 1.6 standard errors); wash volume is 3.1% to 3.3% of volume under every rule, and the ring's gain is unaffected except by the block price. In the uncalibrated default simulator (Appendix D) a minimum resting time of 15 steps does reduce the profit of the same fixed spoofer, so the two simulators disagree on this question; we rely on the Vietnam preset because it matches the observed cancel share, tick and volatility, and the conclusion on order-level frictions holds for that preset only. These rules cost honest investors: a minimum resting time of 15 steps raises the spread by 7% and volatility by 21%, and a cancel fee of 2 raises volatility by 24%. A tick twice as large widens

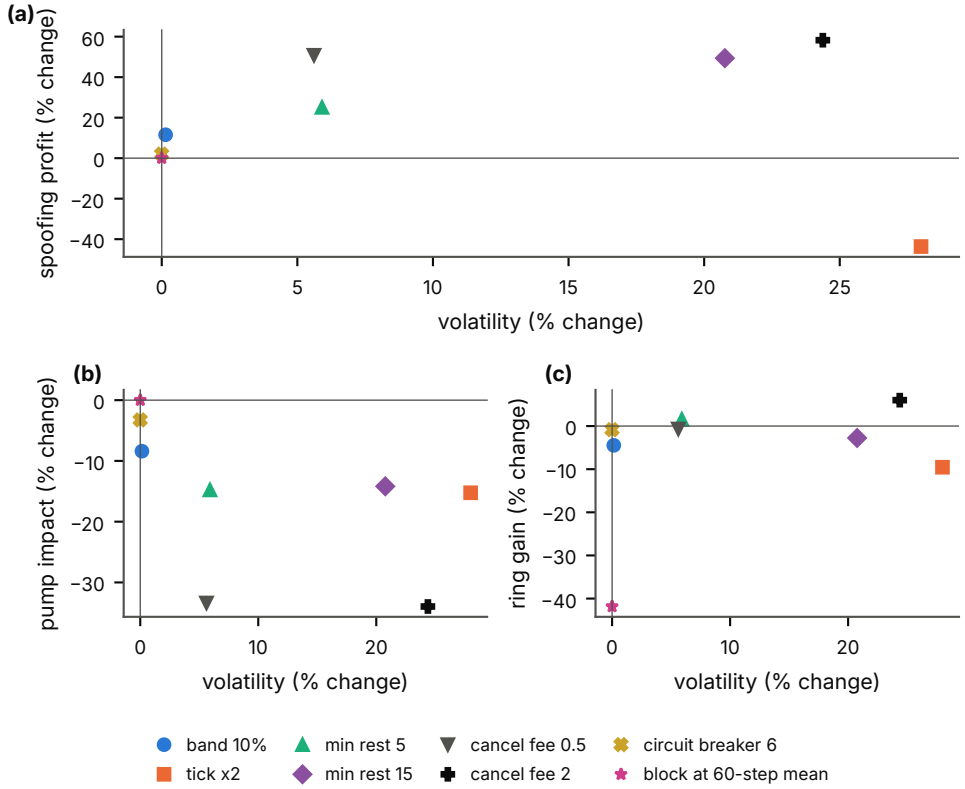


Fig. 11 Effect on manipulators against cost to honest participants, first calibrated market. Each point is an order-level rule or band setting; the horizontal axis is the change of the honest-only volatility, the vertical axis the change in (a) spoofing profit, (b) pump impact and (c) ring gain, relative to the market with a 7% band and no other rule. Manipulators do not adapt

the spread by 35% and cuts depth by 10% while lowering spoofing profit (491 against 872). The block price rule has no effect on honest trading in this experiment because honest agents do not trade blocks; its cost is the gap between the last price and the trailing mean, 0.17% of price at 60 steps and 0.50% at 240 steps in the first preset.

6.8 Checking a lever against reality: the 2016 tick-size reform

The tick size is the one lever in the simulator for which a real natural experiment is available in public data. On 12 September 2016 the Ho Chi Minh Stock Exchange reduced its minimum tick (to 10, 50 and 100 VND below 10,000, between 10,000 and 49,950, and above 50,000 VND), (the tiers are from broker rule summaries, Appendix B; Vo and Doan (2023) study this change and find lower trading costs), and we use HNX and UPCoM stocks as controls on the understanding that it applied to HOSE only. Using daily open, high, low, close and volume, we compare 282 HOSE stocks with 489 control stocks over the 40 trading days before and the 40 days from

policy	spread	depth	volatility	retail cost	volume	spoof impact	pump impact	wash share	ring gain
band 7% (today)	+0%	+0%	+0%	+0%	+0%	0.11	0.47	0.033	34k
band 10%	-0%	-0%	+0%	-2%	-0%	0.14	0.43	0.033	32k
tick x2	+35%	-9%	+28%	-16%	+7%	0.21	0.40	0.031	30k
min rest 5	+2%	-1%	+6%	-5%	-0%	0.16	0.40	0.033	34k
min rest 15	+7%	-3%	+21%	-12%	+1%	0.26	0.40	0.032	33k
cancel fee 0.5	+2%	-2%	+6%	-5%	+1%	0.18	0.31	0.032	33k
cancel fee 2	+8%	-6%	+24%	-5%	+5%	0.20	0.31	0.031	36k
circuit breaker 6	-0%	-0%	-0%	+1%	-0%	0.15	0.45	0.033	33k
block at 60-step mean	+0%	+0%	+0%	+0%	+0%	0.11	0.47	0.033	20k

Table 14 First calibrated market, 10 honest-only and 24 manipulated episodes per policy. Spread, depth, volatility, retail trading cost and volume are relative to the band-7% market; spoofing and pump impact are in ticks; "wash share" is the share of volume traded between colluding accounts; ring gain is in tick-lots. Manipulators do not adapt. Retail cost includes fees paid; it falls under larger ticks and fees because simulated noise traders mostly post passive orders and earn the spread.

one week after the reform. Outcomes are the Corwin–Schultz high-low spread estimator, the daily range, the absolute return, the share of zero-return days and log volume. We could not establish the old tick from our sources (one summary gave 500 VND for all prices, which cannot be right given the small observed spread change), so we compare with simulated reductions of the tick by factors of 2, 5 and 10.

Table 15 and Figure 12 show the result. The share of zero-return days fell by 7.1 points relative to controls (95% interval −9.2 to −5.0), by 8.3 points with 20-day windows and by 6.7 with 60-day windows; the spread estimate fell by 0.06 points (8% of its pre-reform level; interval −0.13 to 0.01), and by 0.07 points with 60-day windows. Because one date and a bootstrap over stocks ignore common shocks, we also compute the same estimate at 57 placebo dates every ten trading days in 2015–2018 whose windows do not overlap the reform's (Figure 13): the fall in the share of zero-return days is 4.3 standard deviations below the placebo mean and no placebo date is as extreme (placebo range −3.1 to +4.1 points), and a permutation of the exchange labels across stocks at the true date gives the same answer ($p < 0.001$ in 5000 draws); the spread estimate, by contrast, is at the 12th percentile of the placebo distribution (two-sided placebo $p = 0.26$, permutation $p = 0.08$) and the daily range is not distinguishable from placebo variation at all. The weekly differences of the spread before the reform have a standard deviation of 0.08 points and no trend ($t = -0.8$), the post-reform mean difference is 0.8 of that standard deviation. Of the 389 HOSE stocks and 926 controls with data, 282 and 489 were listed before the start of the pre-reform window and enter the estimates. By price tercile the fall in the spread estimate is concentrated in low-priced stocks (−0.18 points, interval −0.32 to −0.03, against a pre-reform level of 1.0%). The simulator predicts a fall of the share of zero-return days by 38% for a halving of the tick, 43% for a five-fold cut and 78% for a ten-fold cut (real: 24%), and of the spread estimate by 21%, 65% and 85% (real: 8%). The signs agree, in line with the finding of lower trading costs by Vo and Doan (2023) on intraday data; the real effect is of the order of the simulator's prediction for a halving of the tick and much smaller than for a five- or ten-fold cut. Prices in the panel are adjusted, so price bands are approximate, and the design cannot separate the reform from other changes on HOSE in the same weeks.

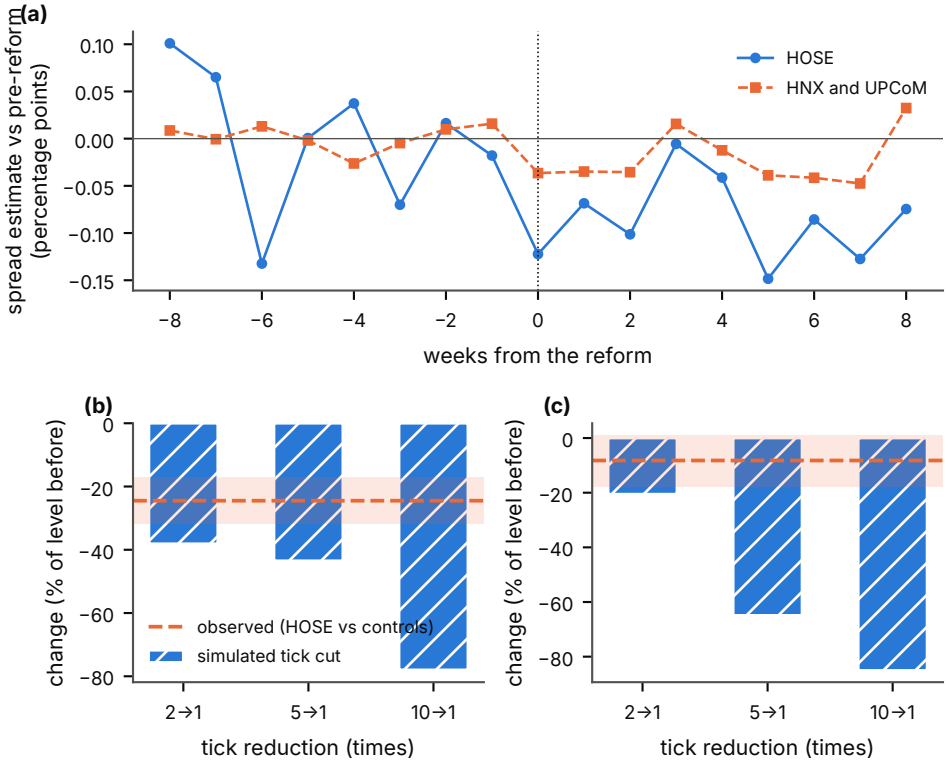


Fig. 12 The 2016 tick reform. (a) Weekly change of the Corwin–Schultz spread estimate relative to the 40 days before the reform for 282 HOSE stocks and 489 controls. (b), (c) Simulated change after a tick reduction by a factor of 2, 5 and 10 (bars) against the observed difference-in-differences (dashed line, shaded 95% interval), as a percentage of the level before, for the share of zero-return days (b) and the spread estimate (c)

outcome	real DiD (95% interval)	real, relative to HOSE pre-period level	sim: tick 2 → 1	5 → 1	10 → 1
CS spread (%)	-0.061 [-0.133, 0.008]	-8%	-21%	-65%	-85%
daily range (%)	-0.067 [-0.213, 0.116]	-2%	-7%	-31%	-50%
—return— (%)	-0.141 [-0.270, 0.012]	-8%	-3%	-35%	-44%
zero-return share	-0.071 [-0.092, -0.050]	-24%	-38%	-43%	-78%
log volume	-0.000 [-0.153, 0.172]	–	–	–	–

Table 15 Tick-size reform: real difference-in-differences against the simulator's prediction for a reduction of the tick by a factor of 2, 5 or 10. Changes relative to the pre-reform level.

6.9 Summary of the policy evidence

Table 16 lists, for each lever, what the simulated markets and the real data show.

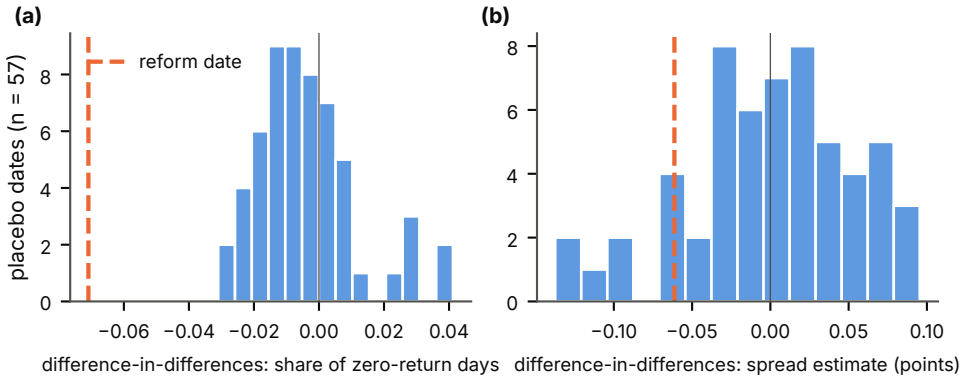


Fig. 13 Placebo distribution of the difference-in-differences (HOSE minus controls, 40+40-day windows) at 57 dates in 2015–2018 away from the reform, for the share of zero-return days (a) and the spread estimate (b); the dashed line is the estimate at the reform date

lever	against rings	against spoofing, pumps, wash	cost to honest investors
inspection with a fine proportional to the gain	deters with enough capacity (three accounts per window: 5 of 6 markets in each of the two families, 2 of 4 without trending); capacity falls by a factor of 2.4 when the fine multiple doubles	not tested	fine paid by manipulators only; inspection effort
off-book block priced at a trailing mean	removes 58–61% of the block value at 60 steps and 87–98% at one day; total gain falls to 28–44% of its value at one day (block of 3000 lots; Appendix D)	not applicable	0.17%–0.50% of price on block trades
daily price band 7% or 10%	no effect in the markets tested	not tested	none observed in honest markets
minimum resting time, cancel fee	no effect	no reduction; spoofing profit and impact not lower	volatility +6% to +24%, spread up to +8%
tick size	not tested for rings	larger tick lowers spoofing profit	spread +35% for a doubled tick; real tick cut improved liquidity

Table 16 Policy evidence in this paper (simulated markets calibrated to a HOSE stock unless stated; the tick row also uses real data).

7 Experiments III: real data

7.1 Real account-level order flow

Every result above is for simulated accounts. To see what the statistic does on real accounts we use two public data sets from a venue with an account identifier on every record, the Hyperliquid derivatives exchange. They are far from the Vietnamese market (a crypto perpetual-futures venue, dominated by automated accounts, with pseudonymous addresses that one operator may use in several copies), so the results are a stress test of the statistic, not an estimate for Vietnam. There are no manipulation labels.

data	accounts	account-windows	unsigned, $p \leq 0.01$	signed, $p \leq 0.01$	unsigned, $p \leq 0.05$	signed, $p \leq 0.05$
fills, BTC	all with ≥ 3	3765	0.156	0.000	0.298	0.000
fills, ETH	all with ≥ 3	1930	0.166	0.000	0.308	0.000
fills, HYPE	all with ≥ 3	3732	0.151	0.000	0.290	0.001
order submissions, SOL	3–9 submissions	843	0.075	0.066	0.198	0.206
order submissions, SOL	10–99	645	0.332	0.326	0.457	0.437
order submissions, SOL	≥ 100	682	0.655	0.518	0.723	0.594
order submissions, SOL	all with ≥ 3	2170	0.334	0.285	0.440	0.397
order submissions, SOL	both sides	1166	0.525	0.436	0.624	0.534

Table 17 Share of accounts flagged on real account-level flow (accounts with at least three records in the window). Under the null of Proposition 1 the share would be at most the level

Data.

Order submissions: the Level-4 order data of Albers et al. (2026) (Zenodo record 18184441, CC BY 4.0), first member of the December 2025 file for the SOL perpetual, 577,918 orders submitted by 1,909 accounts in 23.6 minutes of 1 December 2025; we use one record per order identifier and the side of the order, windows of 300 bins of 0.5 seconds (150 seconds; nine windows with 351 to 715 active accounts and 61,000 orders on average). *Fills*: a public table of fills (account address, side, millisecond time) of a 91-minute slice of 5 June 2026 for the perpetual contracts HYPE, BTC and ETH, windows of 300 bins of one second (18 windows). Accounts with fewer than three records in a window are not tested.

Calibration on real flow.

Table 17 gives the share of accounts with small p-values. On submissions, 28.5% of accounts with at least three orders in a window have $p \leq 0.01$ for the signed statistic (33.4% unsigned), against a nominal 1%, and the share rises with activity: 6.6% for accounts with 3 to 9 orders, 32.6% for 10 to 99, 51.8% for 100 or more; for accounts that trade on both sides it is 43.6%. Netting buys against sells barely lowers the rate, unlike in the simulator: real market makers do not re-quote symmetrically but shift both quotes in the direction of the last price move, so their net flows are positively correlated (Proposition 4 covers the symmetric case only). The exchangeability condition of Proposition 1 is violated for a large share of the active accounts of this venue: they react to the same signals. The aggregate flow of the other accounts is also dominated by a few loud accounts: its variance per bin, divided by the median variance of a quiet account (3 to 15 orders in the window), gives an effective number of accounts $N_{\text{eff}} \approx 10^6$ (median over the nine windows, Section 4.3) against a median of 468 active accounts per window (mean 508; about 240 have at least three orders), so that a ring of quiet accounts is nearly invisible unless it is loud. On fills the pattern is mechanical and different: the signed statistic flags almost nothing (0.0–0.03%) and the unsigned one 15–17%, because each trade appears twice, as the buy of one account and the sell of its counterparty in the same millisecond, so that the aggregate of the others contains the mirror image of every account's own flow. The test must be applied to submissions, not to executed trades.

A ring planted in real flow.

To measure power against the real background, we choose $K = 12$ quiet real accounts (3 to 15 orders in the window) and add to each, at ten common bins of the window

sweep	N	K	orders per push	pushes	unsigned @5	signed @1	signed @5	signed @10	members with $p \leq 0.01$
orders per push	468	12	0	10	0.00	0.00	0.00	0.00	0.09
orders per push	468	12	3	10	0.00	0.00	0.00	0.00	0.03
orders per push	468	12	10	10	0.00	0.00	0.00	0.00	0.06
orders per push	468	12	30	10	0.00	0.00	0.00	0.00	0.41
orders per push	468	12	100	10	0.87	0.59	0.80	0.89	1.00
orders per push	468	12	300	10	1.00	1.00	1.00	1.00	1.00
accounts	100	12	100	10	1.00	1.00	1.00	1.00	1.00
accounts	200	12	100	10	1.00	1.00	1.00	1.00	1.00
accounts	400	12	100	10	0.91	0.78	0.91	0.96	1.00
accounts	468	12	100	10	0.82	0.65	0.79	0.91	1.00
ring size	468	4	100	10	0.00	0.00	0.00	0.00	0.38
ring size	468	8	100	10	0.37	0.19	0.26	0.29	0.91
ring size	468	16	100	10	1.00	1.00	1.00	1.00	1.00
ring size	468	32	100	10	1.00	0.99	1.00	1.00	1.00
pushes per window	468	12	100	3	0.37	0.17	0.22	0.24	0.85
pushes per window	468	12	100	10	0.83	0.64	0.79	0.86	1.00
pushes per window	468	12	100	30	1.00	1.00	1.00	1.00	1.00

Table 18 Planted ring among 12 quiet real accounts: catch@B over 180 trials (nine windows, 20 draws), ranking by the signed statistic (unsigned at 5 for comparison), and share of members with $p \leq 0.01$. The sweep column names the factor that varies; the others are at $N = 468$, $K = 12$, 100 orders per push and 10 pushes

(each member jittered by at most one bin), q buy orders; the other accounts' flow is unchanged. Table 18 and Figure 14 give catch@B over 180 trials (nine windows, 20 draws each). With no planting ($q = 0$) none of the chosen accounts reaches the top five. A ring of three orders per push is undetectable; at 30 orders per member and push (5.9% of the orders of the window in total) it is in the top ten in none of the 180 trials although 41% of its members have $p \leq 0.01$, because more than a quarter of all accounts have $p \leq 0.01$ for natural reasons and the ranking is filled by loud accounts; at 100 orders (19.5% of the flow of the window) it is in the top five in 80% of the trials (signed statistic), and at 300 in all. The dependence on the design is steep and in the direction of Proposition 3: catch@5 falls from 1.00 to 0.79 as the number of accounts rises from 100 to 468 (by dropping real accounts at random), it is 0 for a ring of four, 0.26 for eight and 1.00 for 16 accounts, and 0.22, 0.79 and 1.00 for three, ten and thirty pushes per window. A ring of 12 retail accounts that submits a few orders at each of ten moments would not be found in this venue by this statistic; a ring that makes up a fifth of the flow of a stock-window would be. Whether a Vietnamese stock has a background as loud as this one is the empirical question account-level data would settle, and the planted-ring audit is the way to measure it (Section 8).

7.2 Documented enforcement cases on daily prices

Data and protocol.

Ten decisions of the State Securities Commission with exact dates of the conduct were collected from press reports quoting the decisions (Appendix A; the decision documents were not opened). For every stock and date, using only data up to that date, we compute market-adjusted abnormal return, volume surge against the stock's own history, a round-trip statistic, the position of the close in the daily range, the count of days with moves of at least 6.5%, and range expansion; stocks are ranked against each other on each date. The protocol, fixed before the cases were scored, is: a case stock is scored by the highest rank it reaches on any date in its period; two nulls

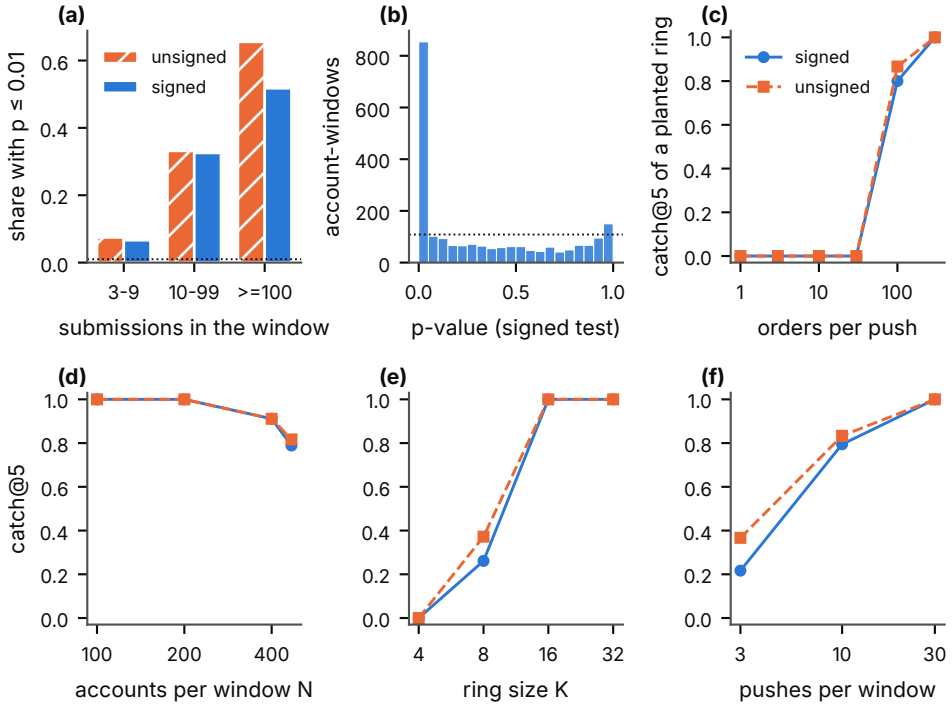


Fig. 14 The coincidence test on real order submissions (SOL perpetual, 23.6 minutes). (a) Share of accounts with $p \leq 0.01$ by number of submissions in the window; dotted: nominal 1%. (b) Histogram of signed-test p-values of accounts with at least three submissions; dotted: uniform. (c)–(f) catch@5 of a ring of 12 quiet real accounts planted at ten common moments per window, signed (solid) and unsigned (dashed), against the number of orders per member and push (c), the number of accounts (d), the ring size (e) and the number of pushes per window (f); unless varied, $q = 100$ orders per push and the full set of accounts

are used, 300 random stocks over the same dates and 300 non-overlapping periods of the same length of the same stock; and we count the cases in which the stock is among the B highest of its date at least once.

Results.

Table 19 gives the results. The market-adjusted abnormal return alone places 8 of the 10 cases in the daily top ten of about 1,360 scored stocks at some point of their period, against 14% for random stocks over the same dates, and a case stock beats a random stock with probability 0.89 and the same stock at other times with probability 0.87. Pooled over the ten cases, the number of cases that reach the top ten is far above what the per-case chance levels imply (8 hits against 1.4 expected, exact Poisson-binomial $p < 0.001$; 9 against 2.3 at 10 days and 3 against 0.7, $p = 0.02$, at 60 days), and Fisher's combination of the per-case p-values against random stocks gives $p < 0.001$ at all three window lengths. These are not stocks that rank high at all times. Combining six features, or an isolation forest on them, is no better than the single feature. Volume

detector	superiority	top 10	chance	top 20	chance	top 50	chance
market-adjusted abnormal return	0.89	8/10	14%	10/10	29%	10/10	54%
combined six-feature score	0.79	6/10	22%	6/10	34%	10/10	57%
isolation forest, same features	0.79	7/10	20%	8/10	31%	9/10	53%
round trip (run-up, retracement)	0.70	9/10	75%	10/10	81%	10/10	90%
volume surge	0.33	0/10	3%	0/10	5%	0/10	13%

Table 19 Ten documented cases scored against about 1,360 stocks (20-day window). Superiority is the probability that a case stock beats a random stock (0.5 is chance); "top B" counts cases in which the stock reached the daily top-B list at some date of its period; "chance" is the same share for random stocks.

window (days)	detector	superiority vs random stocks	superiority vs same stock	cases in top 10 / 10
10	abnormal return	0.87	0.78	9
10	combined score	0.69	0.70	5
10	volume surge	0.33	0.88	0
20	abnormal return	0.89	0.87	8
20	combined score	0.79	0.86	6
20	volume surge	0.33	0.90	0
60	abnormal return	0.85	0.89	3
60	combined score	0.77	0.88	3
60	volume surge	0.28	0.79	0

Table 20 Sensitivity to the window length. Superiority against random stocks and against the same stock at other times; number of cases in the daily top ten.

surge behaves differently in the two comparisons: against the same stock at other times it is informative (superiority 0.90, five of ten cases with $p < 0.05$), against other stocks it is not (0.33), because other stocks have larger surges at some time; we therefore claim nothing new for the market-level detector. Table 20 shows that the superiority of the abnormal-return score is stable across window lengths of 10, 20 and 60 days (over random stocks 0.87, 0.89, 0.85; over the same stock 0.78, 0.87, 0.89), but the number of cases in the daily top ten is not: 9, 8 and 3 of 10, against chance levels for random stocks of 23%, 14% and 7%, and all ten cases are in the top 50 at 60 days (chance 32%). The headline count of 8 of 10 is thus for the 20-day window and for a criterion that asks only that the stock reaches the list on at least one day of a period of several months. The result says little about precision: periods are long (83 to 587 trading days), only ten cases are documented among thousands of price run-ups, and an unlabelled run-up is not an innocent one.

Event time.

Figure 15 aligns the ten cases at the first day of their documented period. The cumulative market-adjusted return of the case stocks is flat before the period and rises through it (mean +0.29 twenty days after the start and +0.42 after sixty), the mean percentile of the abnormal-return score rises from 0.39 sixty days before to 0.78 twenty days after, and the percentile of the volume-surge score does not rise consistently (0.34 at day −60, 0.41 at day 20, 0.28 at day 60), which is the reading of Table 19 in time: the price runs up on ordinary volume.

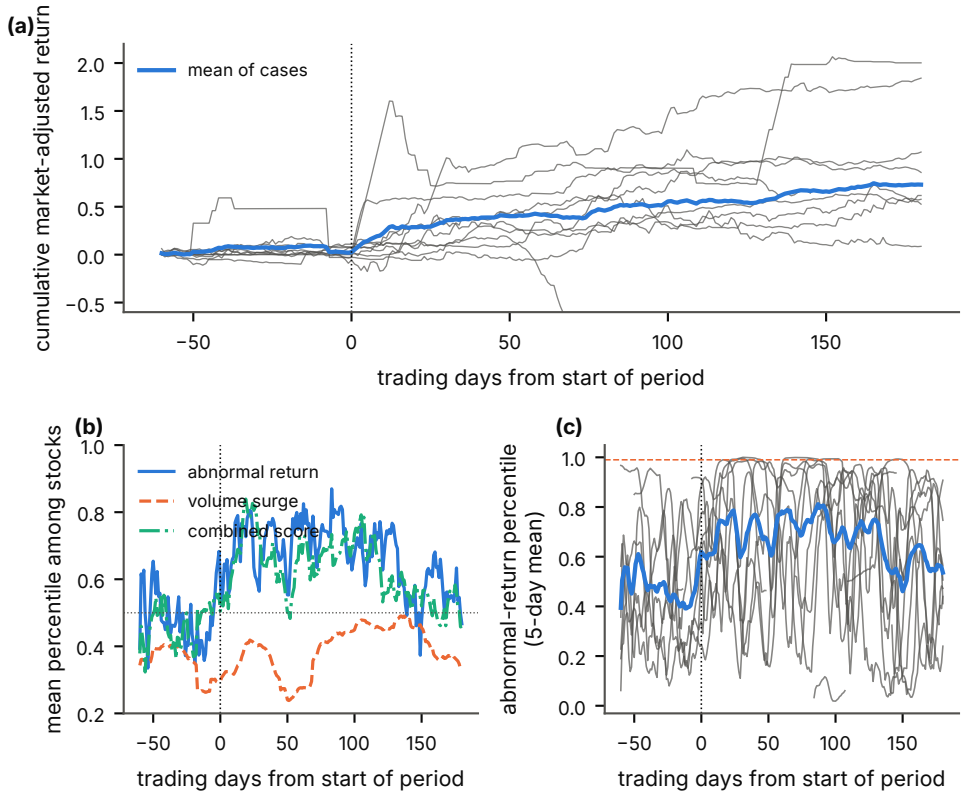


Fig. 15 The ten documented cases in event time (day 0 is the first day of the documented period). (a) Cumulative market-adjusted log return since day −60: individual cases (grey) and mean (blue). (b) Mean cross-sectional percentile among all stocks of three scores computed from data up to each day (20-day windows). (c) Percentile of the abnormal-return score, 5-day moving average, individual cases (grey) and mean (blue); dashed line: the 99th percentile

Case stocks and the daily band.

Table 21 compares the case stocks with the market and counts days within 0.5 points of the daily ceiling. The stocks are ordinary (median daily volatility 1.75% in the year before against 2.46% for HOSE stocks, over the nine case stocks with price history before the period), and the band is reached in their manipulation periods: five ceiling days at the median of the nine, and a run of 12 days for HCI, whose exchange is not in our metadata and whose band we assumed to be 7%. This differs from the simulated ring, whose price displacement of about 0.5% never reaches the band. The band could matter for a manipulator who ratchets the price over days; our simulated rings do not, so the finding that the band has no effect is a statement about rings of the size we simulate, not about all manipulation.

stock	exchange	days	pre-period vol. (%)	vol. in period (%)	zero-return share	log price change (×100)	ceiling days	longest run
FIR	HOSE	109	1.63	1.84	0.02	+38	0	0
SJS	HOSE	192	2.32	2.34	0.13	+49	10	3
PDR	HOSE	83	1.94	3.84	0.07	-119	4	3
PSH	unknown	319	1.75	3.45	0.45	-12	26	3
AGG	HOSE	514	1.29	1.90	0.13	+35	7	1
CRC	HOSE	106	1.36	2.65	0.18	+73	5	1
GKM	HNX	125	1.62	3.77	0.32	+157	1	1
PAS	UPCOM	491	3.37	3.54	0.10	-87	0	0
HCI	unknown	146	6.78	6.17	0.81	+183	21	12

Table 21 Case stocks: daily volatility and zero-return share in the year before the period, volatility during it, price change over the period, days within 0.5 points of the daily ceiling and the longest run of such days.

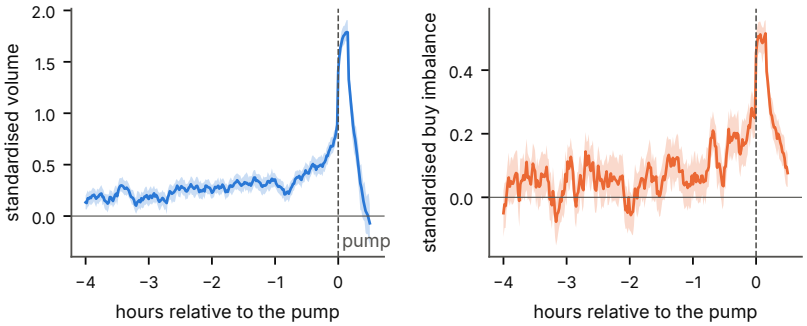


Fig. 16 Standardised volume and buy imbalance of 701 pump events around the pump time (mean and 95% bootstrap interval over events)

7.3 Telegram-organised pump events on tick data

We use a public archive of 701 pump events in cryptocurrency (not Vietnamese) markets, announced on Telegram, the exchange of each event not being identified in our files, with trades from about 24 hours before the pump time to shortly after (Appendix B). The files hold side, price and amount, with no account identifiers, so the coincidence test cannot be applied. For each event, standardised volume and buy imbalance over rolling ten-minute windows are measured against the same coin's first 20 hours. Figure 16 shows accumulation: mean standardised volume rises from 0.19 (4 to 3 hours before) to 0.71 (last ten minutes), and buy imbalance from 0.03 to 0.23, with intervals that exclude zero in the later bins, and both jump at the pump. This supports the accumulate–push structure of the simulated ring. It does not give early warning: with the alarm rate in the background set to 0.05 per coin-hour, only 6.0% of events raise an alarm in the last hour before the pump (8.3% at 0.1 and 16.4% at 0.25 per coin-hour; median lead 7 to 15 minutes when it fires).

8 Discussion: what this means for a regulator

The evidence is simulated for the questions on rules and real for some of the questions on data, and the two do not overlap on accounts. We therefore separate what a

regulator could act on now, what it should do to learn more before acting, and what should not be concluded.

8.1 Findings that can guide rules now

Price off-book blocks from a trailing mean.

Among the price and order rules we tested this is the only one that cuts the ring's gain through a mechanism that can be written down and checked: a reference price that averages the last L prices credits a push only for increments that are recent and discounts them linearly with age (Proposition 6). In the simulated markets a one-day reference removes 87% (index level) to 98% (stock level) of the value of the block and leaves the ring between a tenth and a half of its total gain (Section 6.5); this holds for a block of the assumed size, and a ring whose gain is mostly trading profit (blocks ten times smaller) keeps more than 80% of it; a ring can recover the value only by holding the price for about $L - a/2$ steps, which is itself the conduct to deter. Its cost to honest trading is the gap between the last price and the trailing mean on block trades, 0.2% to 0.5% of the price in our first preset, which we have not measured on real negotiated trades. One of the ten documented cases (PSH) was carried out on a negotiated system, and the rule applies where the ring's gain is realised off the order book; it does nothing against manipulation that profits on the book. We would pilot it first on a small set of stocks and compare negotiated-trade prices with the trailing mean to measure the cost before it binds.

Inspection capacity and the fine are exchangeable.

With a fine proportional to the gain, deterrence requires only that the probability of being caught exceed $1/m$ (Proposition 5). Under the fitted power law, doubling the fine multiple from five to ten cuts the capacity needed to deter a small ring by a factor of about 2.4 (3.0 to 1.2 inspected accounts per window at 600 active accounts; Table 13). The Vietnamese caps (five times the gain for individuals, ten for organisations, as reported by the press) therefore matter for how many accounts the inspectorate must examine, and an increase in the cap for individuals would be worth about as much as a 2.4-fold increase in capacity under our model. The empirical counterpart is the finding of Comerton-Forde and Putniņš (2014) that the regulator's budget has a strong effect on both manipulation and detection. The arithmetic of a workload is simple: B inspected accounts per window, S stocks and W windows per year give BSW account reviews a year; with $B = 3$, an illustrative 400 stocks and about 200 windows (a window is about 300 minutes), that is 240,000 automated reviews a year, of which only the highest-ranked need human attention. Because the saving from screening shrinks as the number of accounts per stock grows (Table 13: saving of 3.7 at 600 accounts and 2.0 at 2000), capacity should be allocated first to stocks with few active accounts, where screening is most informative.

Do not use order-level frictions or the band against rings.

Minimum resting times and cancel fees did not reduce spoofing, pump-and-dump or wash trading in the calibrated market and cost honest participants 7–8% in spread

option	effect found	evidence	cost or risk	first step
price off-book blocks from a trailing mean ($L \approx$ one day)	removes 87–98% of the block value; total ring gain falls to 28–44% (for the assumed block of 3000 lots)	simulated, two families, with an exact ex-post check and a closed-form law	0.2–0.5% gap to the last price on block trades (not measured on real trades)	compare negotiated-trade prices with trailing means on a few stocks
higher fine multiple, or more inspection capacity	doubling the multiple from 5 to 10 cuts the capacity needed to deter by a factor of 2.4	simulated; closed form fitted to the agent-based market at about 100 accounts and extrapolated with $N^{-1/2}$; weaker in markets with more accounts (2 of 4 without trending)	legal change; staff; false flags	compute the workload BSW; plan capacity by number of accounts per stock
screening in shadow mode on account-level data	measures the share of accounts flagged and the effective number of accounts	real order flow elsewhere shows the null fails; none for Vietnam	data governance; interpretation of flags	planted-ring audit on the regulator's own history
no minimum resting time, cancel fee or larger tick as anti-manipulation tools	avoids +7–8% spread, +21–24% volatility, +35% spread	simulated; tick reform on real data	none	none
change of the daily band	none for rings of the simulated size	simulated; case stocks reach the ceiling	unknown for rings that ratchet the price	study case stocks with account data

Table 22 Options for the regulator, the effect found, the strength of the evidence, cost or risk, and a first step.

and 21–24% in volatility at the strictest settings; a larger tick widened the spread by 35% (Figure 11). Widening the daily band to 10% or 15% changed nothing for rings of the simulated size. These results are conditional on the simulator: the ring moves the price by about 0.5% and never reaches the band, whereas the case stocks reached the ceiling for a median of five days and, for one stock, 12 in a row (the band of that stock is assumed) (Table 21). Whether the band constrains a ring that ratchets the price over days is a question for account-level data.

On the 2016 tick reform.

On the data we have, the reduction of the tick on HOSE in 2016 lowered the share of zero-return days by 7.1 points relative to control exchanges, a change that none of 57 placebo dates in 2015–2018 matches (4.3 placebo standard deviations), and the estimated spread by about 8% (the interval includes zero, and the estimate lies at the 12th percentile of the placebo distribution, so it is not distinguishable from the variation of the spread difference at other dates), in the direction the simulator predicts and consistent with lower trading costs found on intraday data (Vo and Doan 2023). The evidence is one reform date with 282 treated and 489 control stocks, so inference over time rests on the placebo dates: the spread estimate depends on the window (a fall of 0.03 points over 20 days with an interval that includes zero, 0.06 over 40 days, 0.07 over 60 days), and we cannot rule out other changes on HOSE in the same weeks. We therefore read it as evidence on liquidity, not as a reason to keep a given tick. Tick size is a liquidity instrument and not a manipulation instrument: we tested it against a ring only with a non-adaptive ring in one market, and the doubling of the tick in the simulator costs honest participants a third of the spread.

Table 22 puts the options side by side with the strength of the evidence behind each and the first step a regulator could take.

8.2 What a regulator should do to learn more before relying on the test

Run the test in shadow mode on account-level data.

The one thing that would change our conclusions most is access to account-level order data, which only the exchange, the depository and the commission hold. On such data the first deliverable is not a ranking of accounts but the calibration check of Section 7.1: the share of accounts with small p-values by activity tier, which measures how far the exchangeability condition is violated in that market.

Audit the detector with planted rings in the regulator's own history.

The planted-ring experiment on real order flow (Section 7.1) is a procedure that does not need labels: inject synthetic coordinated orders into a historical window, vary the size, activity and number of accounts, and measure catch@B against the real background of the stock in question. It yields the capacity B needed per stock for a target detection probability and reveals stocks in which the background of automated accounts makes the test uninformative. Because it uses the venue's own flow, it also estimates s_0 of (8) for that venue.

Treat a flag as a screening signal, not as evidence.

The test controls false alarms per account only under an exchangeability condition that real order flow violates (Section 7.1); with ranking the guarantee is lost (Proposition 2); and in an algorithm-dominated venue the highest-ranked accounts are likely to be honest bots that react to the same signals. The ranking is therefore an input to human review, with a documented route to appeal, not a basis for sanctions. In the simulator a fund that splits orders over sub-accounts, which is coordinated in time but is not a manipulator, is flagged at six times the nominal rate, and a regulator that acts on flags must expect to meet such cases.

8.3 Risks and unintended effects

Who bears the cost.

The block-price rule falls on those who trade off the order book; in the simulator it has no effect on honest trading because honest agents do not trade blocks, and its cost on real negotiated trades is not measured. Order-level frictions raise volatility and spreads for all participants (Table 14); the simulated retail trading cost falls under larger ticks and fees because simulated noise traders mostly post passive orders, which we regard as an artefact, so we do not rank who pays most. The fine falls on manipulators. Inspection has a burden of its own: with $B = 3$ inspected accounts in each of 600 active accounts per window and 200 windows a year, an account chosen at random would be inspected $3 \cdot 200/600 = 1$ time a year, so that every honest account faces about one inspection a year if the ranking were uninformative, and in practice more for loud and automated accounts, which the ranking favours (Section 7.1). The burden is small in effort and potentially large in reputation, which is a reason to keep the first inspection a data request and not a sanction.

Displacement.

Under inspection the best-responding ring does not stop; it shrinks, to about 3% of its unconstrained gain in the stock-level family. Deterrence of large rings may push manipulation toward smaller operations that fall below the inspection threshold, which is the right objective if harm is proportional to size and the wrong one if many small rings together do equal harm. The law of Proposition 3 says where the threshold lies: a ring of k co-active accounts is detectable if $k - 1 \gtrsim c\sqrt{N}$, so the threshold rises with the square root of the number of accounts.

Evasion once the rule is known.

A published rule invites a response. Our strategy space includes jitter (spreading members' actions in time) and cross trades, but the ring cannot, for example, trade across several stocks, split its activity over several windows with a different group of accounts, or imitate the order flow of an honest group. Randomising the inspected set among the top-ranked accounts, and keeping the exact statistic confidential, are measures we have not tested.

Privacy and governance.

Account-level surveillance data are sensitive. A shadow-mode pilot should run inside the exchange or the commission, with the output reduced to ranked pseudonymous identifiers, and with the false-flag statistics of the pilot published before any use for enforcement.

8.4 Generalisation

The structure that drives the results is not specific to Vietnam: a market dominated by retail investors, a daily price band, off-book or negotiated trades priced from a reference, administrative fines that are a multiple of the gain, and a regulator with limited inspection capacity. For another market, the quantities to re-estimate are the cancel share, the retail share, the volatility of the index and of a typical stock, the number of active accounts per stock, and the reference-price rule for blocks; the family-of-markets design then gives the distribution of the ring's value over markets that match those moments. The introduction of a new trading system in Vietnam in May 2025 may raise the share of automated accounts; Section 7.1 shows that on a venue dominated by automated accounts the coincidence test flags many accounts and loses power, which is a reason to calibrate it on the new flow before relying on it; we did not vary the share of automated accounts.

8.5 Limitations

No real account-level data from Vietnam have been used, so we do not know whether real rings are separable from honest coordination by this test. The simulator's manipulators and honest groups were written by the same author as the detectors, and a model trained with labels in the simulator beats the label-free test in every scenario. The calibrated markets do not reproduce the spread, the tails or the absence of serial correlation in daily returns (Figure 2; a third family that restores it is analysed in

Section 6.3). Rule facts come from broker summaries and press, and we could not establish the pre-reform tick. The ring search has a small budget and the strategy set was found in the first market; the strategies were not re-optimised in each market of the families, so the reported effects of the rules are lower bounds on what an adaptive ring gains (Proposition 7). The block value of 3000 lots is an assumption; a ring that is caught is assumed to be exposed by one inspected member; and fixed manipulators other than the ring are not best responses. The power law is an approximation fitted to 11 points with two constants. The signed statistic is blind to a ring whose members trade against each other in balance (Section 5.3), which is how four of the ten documented cases are described, and the unsigned statistic that sees it is the one that honest desks defeat. The ring's strategy menu has four elements, and 24 random strategies added in every market did not raise the ring's gain against the band or block pricing and raised the number of markets in which inspection fails to deter from 3 to 4 of 16 (Section 6.4), which is not a proof that the menu is near the optimum; the deterrence results assume that detection is independent of the gain; the rule results come from six markets per family (four without trending), with wide intervals over markets (Section 6.4: block pricing at 240 steps lowers the ring's gain in all 15 markets with a positive baseline, but the median share kept is 0.12 with a bootstrap interval of -0.01 to 0.24); and the supervised model transfers across simulator configurations (Section 5.6) but we do not know whether it transfers to real rings. The real account-level data are pseudonymous addresses on a crypto derivatives venue, and the planted ring is synthetic. The Vietnamese checks are at the level of the stock; the case periods come from press reports; the panel holds only stocks that are still listed; the natural experiment compares HOSE with other exchanges in the same weeks and cannot rule out other differences.

8.6 Next steps

Account-level data from the exchange, or the per-account trade tables in published case files, would allow a direct test of the central claim. A rule based on finer resampling and a false-discovery-rate bound should replace top-B ranking, whose sensitivity falls with the number of accounts. A test conditioned on the common drivers of order flow (price moves, news, activity regimes) is needed for algorithm-dominated markets. The interplay of an adaptive ring with an adaptive regulator is a repeated game that double-oracle methods could solve (McMahan et al. 2003; Lanctot et al. 2017). And the optimal combination of fine, capacity and rules is a design problem that the closed forms of Section 4.5 make tractable.

9 Conclusion

A regulator without labelled cases can screen for coordinated accounts with a test whose false-alarm rate is exact per account under an exchangeability condition, whose power falls like the inverse square root of the number of accounts, and whose failure profile we have measured on simulated honest groups and on real order flow. Against a ring that adapts, pricing off-book blocks at a trailing mean and a sufficient inspection capacity under a fine proportional to the gain remove most of the ring's expected

benefit in simulated markets that match the observable moments of the Ho Chi Minh Stock Exchange (for the block rule, when the block is the main source of the ring's gain), while the daily band does not, and minimum resting times and cancel fees, tested in one market against fixed manipulators, do not and cost honest investors liquidity. The central uncertainty is the one data would settle: whether real rings in a retail-dominated market look like the simulated ones against a background of honest accounts that is as synchronised as the real one.

Appendix A Documented cases considered in Section 7.2 (HNG is listed but not scored)

ticker	period	accounts	decided	type
FIR	2022-01-04 to 2022-06-17	76	SSC 2023-11 and 2024-01	cross trading among own accounts
SJS	2023-04-10 to 2024-01-10	26	SSC 2026-01-08	continuous buying and selling
PDR	2022-08-15 to 2022-12-09	164	SSC 2025-03-21	cross trading among own accounts
PSH	2021-02-01 to 2022-05-27	26	SSC 2024-05-27	negotiated trades within a group
AGG	2020-12-01 to 2022-12-31	92	SSC 2025-09 (year inferred)	continuous buying and selling
PPT	2022-06-17 to 2024-10-17	19	SSC decision 881	continuous buying and selling
CRC	2025-03-03 to 2025-08-01	10	SSC 2025-12-22	cross trading
GKM	2021-08-02 to 2022-01-28	23	SSC 2023-12-21	fake supply and demand to push the price up
PAS	2021-01-04 to 2022-12-30	19	SSC 2026-02-12	continuous buying and selling
HCI	2019-12-13 to 2020-07-16	5	SSC 2024-05-30	cross matched orders among own accounts
HNG	2015-07-20 to 2016-04-01	42	SSC 2017-08-10	multi-account fake supply and demand

Table 23 Cases with exact dates of the conduct. The periods are from press reports quoting the State Securities Commission decisions (sources are listed in the repository file `data/cases.csv`); the decision documents were not opened. FIR was the subject of two decisions with the same period and is counted once. PSH was traded on the negotiated system.

Appendix B Sources for the calibration and real-data checks

- Tick sizes, daily bands, lot sizes, sessions: broker rule summaries (FPTS for HOSE, a securities-company PDF for HNX); the system of the Korea Exchange (KRX) went live on 5 May 2025. Details and confidence tags: `docs/vietnam_calibration.md` in the repository.
- Fine cap of five times (individuals) and ten times (organisations) the illegal gain: Vietnamese press and legal-news summaries of Decree 156/2020 as amended; not read from the decree.
- Cancel and amendment share of 31.76% of orders (December 2020): press report of a HOSE statement.
- VN30, VN-Index and VN100 bars and the VN30F1M ticks: public Kaggle datasets (keithvo/vnstockdata, khimduong/vn30-market-making).
- Daily prices of 1,551 stocks: public Kaggle dataset (vuthinh/vietnam-stock-market-ohlc-price-data).

- Pump events: public Hugging Face dataset (Go3x3/pump_and_dump_dataset, MIT licence).
- Order submissions with account identifiers: Albers et al. (2026) (Zenodo record 18184441, CC BY 4.0), first member of the December 2025 SOL file of about 880 million records per day over BTC, ETH and SOL; fills with account addresses: a public Hugging Face table of Hyperliquid fills (craftify2221/hyperliquid-fills-raw; the dataset card declares no licence), one row group of 5 June 2026. Both are mirrors or derivatives of the venue's public node data; we use samples of the files, not the full archives.
- Exchange metadata (HOSE, HNX or UPCoM for each ticker): public Hugging Face table of Vietnamese stock symbols (kjhq/Vietnam-Stock-Symbols-and-Metadata).

Appendix C Single-market results

The first version of the policy comparison used one calibrated market (index level, daily volatility 1.14%), with 20 selection and 20 reporting episodes per cell and seven strategies. Table 24 and Figure 17 are from it; they include the daily bands of 10% and 15% and a fine multiple of 10, and they are consistent with the family results in direction (block price: large reduction; inspection: negative net benefit at one and three inspected accounts in that market), but the band of 7% appears to halve the gain in the unrestricted evolution-strategy search of Section 6 (101k without a band, 49k with it) and not in this cross-evaluation, which shows the noise of the search. In this market the single inspected account is caught in 20% of the episodes (gain 14k, standard error 11k over 20 episodes).

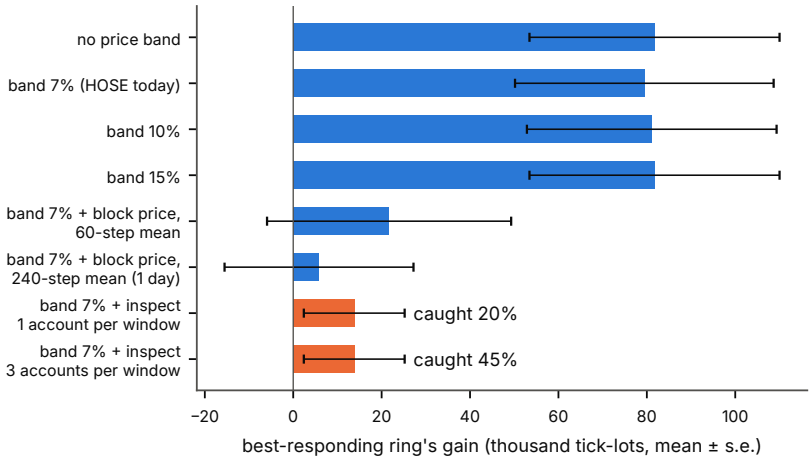


Fig. 17 Single market: gain of the best-responding ring (thousand tick-lots, mean and standard error over 20 episodes)

policy	ring gain (mean, s.e.)	caught	net benefit after fine, $m = 5$	$m = 10$
no price band	81.7k (28.3k)	–	–	–
band 7% (HOSE today)	79.4k (29.3k)	–	–	–
band 10%	81.1k (28.2k)	–	–	–
band 15%	81.7k (28.3k)	–	–	–
band 7% + block at 60-step mean	21.7k (27.6k)	–	–	–
band 7% + block at 240-step mean	5.8k (21.4k)	–	–	–
band 7% + inspect 1 per window	13.8k (11.4k)	20%	-35.6k	-85.1k
band 7% + inspect 3 per window	13.8k (11.4k)	45%	-48.9k	-111.6k

Table 24 Single market: best-responding ring against each policy; net benefit with fines of $m = 5$ and $m = 10$ times the gain.

Appendix D Additional tables

Table 25 gives the moments of each accepted market re-measured over 16 honest episodes, and Table 26 the block-price rule against the assumed block size. Table 27 gives the ablations of the signed test behind Figure 7, and Table 28 the effect of order-level rules on the default, non-adaptive spoofer in the uncalibrated simulator, which the main text does not use.

family	market (draw)	volatility at acceptance (%)	volatility, 16 episodes (%)	cancel share	retail share	spread (ticks)
index level	1	1.08	1.10 (0.06)	0.316	0.711	3.04
index level	5	1.13	1.63 (0.12)	0.262	0.730	3.38
index level	11	1.08	1.26 (0.10)	0.331	0.699	3.07
index level	19	1.17	1.40 (0.10)	0.328	0.670	2.86
index level	20	0.93	1.19 (0.06)	0.271	0.752	3.46
index level	26	1.18	1.22 (0.10)	0.303	0.724	3.27
stock level	14	2.03	1.65 (0.08)	0.277	0.728	3.45
stock level	19	2.19	2.58 (0.17)	0.276	0.663	3.17
stock level	23	2.48	2.06 (0.14)	0.302	0.692	3.35
stock level	39	2.19	2.23 (0.16)	0.277	0.672	3.23
stock level	49	2.42	2.87 (0.23)	0.265	0.695	3.43
stock level	61	1.84	1.81 (0.16)	0.282	0.709	3.28

Table 25 Moments of the accepted markets re-measured over 16 honest episodes of 6000 steps (standard error of the volatility in brackets), against the volatility measured at acceptance on two episodes. Tolerances: volatility 0.9–1.4% (index level) and 1.8–2.8% (stock level), cancel share 0.27–0.37, retail share 0.66–0.85

	index level			stock level		
block size (lots)	gain, last price	kept at $L = 60$	kept at $L = 240$	gain, last price	kept at $L = 60$	kept at $L = 240$
300	42k	0.89	0.83	68k	0.92	0.87
1000	70k	0.74	0.60	101k	0.78	0.64
3000	151k	0.52	0.28	297k	0.68	0.44
10000	435k	0.49	0.19	693k	0.53	0.21

Table 26 Block-price rule against the assumed block size (median over six markets per family): gain with the last price as reference, and the share of it kept when the block is priced at the mean of the last 60 or 240 steps, for a ring that keeps its strategy. Assumed size: 3000 lots

factor	level	catch@5	catch@1	honest flagged	accounts per window
tolerances	(1,)	0.89 [0.85, 0.94]	0.44 [0.38, 0.52]	0.013	100
	(1, 5)	0.88 [0.83, 0.93]	0.55 [0.48, 0.62]	0.013	100
	(1, 5, 15)	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100
	(1, 5, 15, 30)	0.90 [0.86, 0.95]	0.55 [0.49, 0.61]	0.014	100
window length (steps)	150	0.88 [0.83, 0.91]	0.41 [0.34, 0.47]	0.007	81
	300	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100
	600	0.91 [0.84, 0.97]	0.58 [0.46, 0.71]	0.018	108
noise traders (accounts)	80	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100
	200	0.84 [0.76, 0.90]	0.45 [0.34, 0.55]	0.010	206
	400	0.64 [0.55, 0.71]	0.17 [0.10, 0.25]	0.009	373
	800	0.28 [0.24, 0.34]	0.08 [0.03, 0.14]	0.010	708
ring size K	4	0.79 [0.72, 0.85]	0.41 [0.33, 0.49]	0.013	93
	8	0.88 [0.84, 0.94]	0.50 [0.42, 0.60]	0.012	97
	16	0.94 [0.89, 0.97]	0.53 [0.44, 0.60]	0.012	104
	30	0.94 [0.91, 0.97]	0.63 [0.56, 0.69]	0.012	115
ring push probability	0.2	0.46 [0.37, 0.55]	0.13 [0.08, 0.20]	0.012	100
	0.4	0.79 [0.73, 0.87]	0.31 [0.23, 0.39]	0.014	100
	0.8	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100

Table 27 Ablations and sensitivity of the signed test in a clean market (16 episodes per row, 95% bootstrap intervals). "Honest flagged" is the share of honest accounts with $p \leq 0.01$.

policy	spread	volatility	retail cost	spoofing impact	spoofing profit
baseline	2.6 ticks	0.21	2.2	1.07	56
tick ×4	+124%	+50%	−54%	−25%	687
min. rest 5 steps	0%	−4%	+11%	+1%	13
min. rest 15 steps	+3%	+4%	+16%	−17%	−346
cancel fee 2	+1%	−1%	+36%	−5%	128
circuit breaker (6 ticks)	+1%	+2%	−6%	+3%	43

Table 28 Order-level rules against the default (non-adaptive) spoofer, 16 episodes per policy; changes relative to the baseline. Retail cost includes fees paid and falls under larger ticks because simulated noise traders mostly post passive orders and earn the wider spread.

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Statements and Declarations

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Ethics approval. The study uses simulated data and public market data; no human participants or animals were involved. Enforcement cases are identified by stock ticker only, and persons named in the decisions are not named here.

Data availability. All simulated data can be regenerated with the code and fixed random seeds provided with the paper, and the result tables behind every table and figure are in the repository. The real data are public: daily prices of Vietnamese stocks, VN30 index and VN30 futures data, exchange metadata and Telegram pump-and-dump events, from the sources listed in Table 3 and Appendix B. Raw files are not redistributed because of their licences; the case list used in Section 7 is included in the repository. No Vietnamese account-level or other non-public data were used; the account-level order data used for the stress test of Section 7.1 are public (Zenodo record 18184441 and the table of fills cited in Appendix B).

Code availability. The simulator, detectors, experiment scripts, result tables and the case list are in the repository https://github.com/vinhqdang/quantitative_trading.